

Beliefs from Cues

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Abstract

Beliefs often depend on contextual cues that influence which states come to mind. Models of associative sampling capturing this idea give a central role to *similarity*, which is typically unobserved and in principle very high-dimensional. We show theoretically that, despite this potential complexity, the effects of cues depend only on two sufficient statistics—representativeness and resonance—that summarize how similarity distorts beliefs. Further, these statistics are measurable with simple, pairwise similarity judgments, offering a straightforward method of generating and testing quantitative predictions. In a controlled experiment, we deploy this method and show that belief responses to normatively irrelevant cues closely track model predictions. We also show that large language models can approximate human similarity judgments and extend the approach to field settings. Applying the framework to survey data on personal experiences and macroeconomic expectations, we show that similarity-based predictions quantitatively explain much of the heterogeneity in experience effects across a wide range of experiences and expectation domains.

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1 Introduction

Standard models in economics—even those incorporating behavioral frictions—tend to assume that beliefs are a function only of objectively relevant pieces of information, such as an agent’s priors and the likelihood of signals. But there is growing evidence for the importance of *cues*: contextual triggers that shape beliefs by shifting which past experiences or future possibilities come to mind, over and above the objective information (if any) that they may convey. Cueing effects have been documented across many economic domains, including financial investment, consumption, advertising, macroeconomic forecasting, and charitable giving (e.g., [Bordalo, Burro, Gennaioli, et al. 2025](#), [Gennaioli et al. 2024](#), [Charles 2025](#), [Taubinsky et al. 2026](#), [Augenblick et al. in press](#), [Henkel et al. 2026](#)). Theories modeling such forces give a central role to similarity: cues bring to mind items similar to or associated with them, shaping beliefs even when these connections are not normatively relevant ([Bordalo et al. 2021, 2023](#); [Bordalo, Burro, Coffman, et al. 2025](#), [Graeber et al. 2024](#), [Enke et al. 2024](#)).

Despite this progress, challenges remain for adopting these frameworks more broadly. First, similarity is typically unobserved and, at least in principle, extremely high-dimensional: many features may contribute to determining whether two items are associated with each other, and much work in psychology has studied the subtleties of human similarity judgments ([Tversky, 1977](#); [Medin & Schaffer, 1978](#); [Nosofsky, 1992](#)). Second, though some cues are often observed—for example, experiences elicited in a survey, or experimentally manipulated stimuli—inevitably much of people’s contexts is not. In part due to such challenges, existing work has largely tested the predictions of similarity-based models qualitatively (e.g., testing the sign of cueing effects) and in settings where researchers have strong *ex ante* intuitions about the similarity between cues and what they bring to mind. These observations raise an important question: how can we generate sharp quantitative predictions despite the apparent complexity and (partial) unobservability of similarity and cues?

In this paper, we address this question theoretically and experimentally. We start from a simple model where cues bring to mind states of the world similar to them. We show that—despite few assumptions limiting the complexity of similarity or the number of cues—the effect of cues on beliefs reduces to two sufficient statistics: what we call their representativeness and resonance. We show that the theory offers guidance on how these statistics can be estimated from simple pairwise similarity elicitation. We then apply this method to a controlled experiment where we show that participants’ beliefs about an abstract generating process are highly reactive to uninformative cues. These effects are

predictable from estimates of their representativeness and resonance computed using both human and large language model (LLM) similarity judgments. This latter finding raises the possibility of generating automated predictions of cue effects at scale, without collecting bespoke human similarity data. We conclude by showing that this method helps to predict the effect of dozens of experiences on many macroeconomic expectations, including about inflation, the stock market, unemployment, and interest rates.

We begin with a simple theory of beliefs from cues that draws on similarity-based sampling frameworks (e.g., [Kahana 2012](#); [Bordalo et al. 2023](#)). Our agent forms beliefs about a state-dependent variable, such as the return on an investment or how the unemployment rate will evolve. While a standard agent would integrate over every possible state, our agent instead must form her beliefs based on the states that happen to come to mind. The crucial assumption we make is that the probability that a state comes to mind is proportional to its similarity to the agent’s context, which we model as a set of cues. Our main theoretical result is that, despite the potential high dimensionality of similarity, the effect of any cue on beliefs reduces to two sufficient statistics: its *representativeness*—how differentially similar it is to states where the variable takes on a high versus low value¹—and its *resonance*: how similar it is to the state space as a whole. We then show that simple pairwise similarity elicitation suffices to estimate both these statistics and therefore predict the effect of any cue, even without full knowledge of the underlying similarity function.

We then apply this method in a controlled experiment. Participants are tasked with forming beliefs about a simple data-generating process, framed as a “game.” Each game consists of seven rounds that either add or subtract points from its total score. The number of contributed points each round is uniformly and independently drawn from the set $\{+1, +2, +3, +4, -5, -6\}$, framed as rolls of a six-sided die. Participants then form incentivized beliefs regarding the distribution of the final score, which is the sum across all seven rounds. We ask participants the probability that the game ends up with a final score greater than zero as well as the likelihood of games with even vs odd final scores. Note that the large number of states—all sequences of seven die rolls—precludes participants from simply enumerating all possible outcomes, raising the possibility that cues may influence which states come to mind and thereby distort beliefs.

To test for such effects, participants are randomized to either form beliefs absent any

¹When states vary continuously, representativeness is formally defined as the normalized covariance between the similarity of the cue to each state and the value of the variable in that state. Our notion of representativeness formalizes and extends the original notion of representativeness as a judgment of relative similarity [Kahneman & Tversky \(1972\)](#), as well as likelihood-ratio definitions of representativeness as in [Gennaioli & Shleifer \(2010\)](#) and [Bordalo et al. \(2016, 2018\)](#).

additional cue or to do so only after watching an example game. These example games—the cues in our experiment—are objectively uninformative to a standard rational agent: the data-generating process is already fully described, logically pinning down the objective answers, and the example is only one of the hundreds of thousands of equally likely possible outcomes. Thus, the rational benchmark—and indeed the benchmark for any agent who responds only to objectively informative signals (e.g., even one who under- or overreacts to data)—is that these cues should have no effect on beliefs.² After belief elicitation, participants are then asked to make pairwise similarity judgments between example games. We use these similarity data to estimate the resonance and representativeness of cues.

We find that these cues can have large effects on beliefs, which are systematically related to their resonance and representativeness as measured by our similarity data. Moving from a cue that is representative of negative scores to one representative of positive scores increases beliefs about the share of positive outcomes from 39.8% to 58.9% (77% of a standard deviation, $p < 0.001$). Furthermore, cues that depict explicitly impossible states—ones that participants understand do not follow the rules of the game—also have large effects on beliefs (54.6% vs 43.8% for positive vs negative cues, $p < 0.001$) because they are nonetheless similar to possible states.³ We next show that cues with *zero* representativeness—i.e., which are no more similar to one hypothesis than another—fail to move beliefs. In contrast to the large effects that cues have on beliefs about positive vs negative outcomes, beliefs about the share of games that end with even vs odd scores are not affected by cues. Our similarity data predict this null result: participants do not judge even vs odd cues as differentially similar to even outcomes, and therefore these cues are not representative of that hypothesis.

We then structurally estimate the model using average beliefs across 360 randomly generated cues. Despite having only three degrees of freedom, we find that our fitted model closely tracks average beliefs across cues and across questions: it predicts the size and shape of the effects of positive vs negative cues on beliefs about positive outcomes while also correctly predicting no effect of even vs odd cues. Following [Fudenberg et al. \(2022, 2026\)](#), we calculate the completeness and restrictiveness of the model: how well it fits our data relative to a best-case benchmark (completeness) and how poorly it spuriously fits synthetically generated fictitious data (restrictiveness). We find the model achieves 81.8% of the explanatory power of the best-case machine-learning benchmark

²Furthermore, participants are explicitly told that the examples are randomly assigned and should not have any effect on their beliefs, and comprehension questions confirm their understanding of this fact.

³We also elicit suggestive evidence on which outcomes participants felt came to mind while forming beliefs: these process data strongly track treatment effects on beliefs.

while being 97.6% restrictive: that is, the model fits the actual data well while being almost entirely unable to fit synthetic data.

We conclude by exploring how our methodology can be extended beyond controlled experimental settings where human similarity data are readily available. We first explore LLMs as an alternative source of similarity judgments. Mirroring our human elicitation procedure, we provide an LLM with the experimental setup and ask it to rate the similarity of cue–state pairs. We then use these ratings to estimate each cue’s representativeness and resonance. We find that LLM-based similarity measures predict the effects of cues on beliefs almost as well as human data (75.0% completeness, compared to 81.8% for human similarity data). This suggests that LLMs may provide a useful, low-cost alternative when human similarity data are unavailable.⁴

We then apply this methodology to a field setting, where we investigate the link between personal experiences and macroeconomic expectations (Malmendier, 2021; Taubinsky et al., 2026; Kuchler & Zafar, 2019; Bordalo, Gennaioli, et al., 2026). Using data from the NY Fed’s Survey of Consumer Expectations (SCE), we elicit LLM similarity judgments between respondents’ personal experiences and past macroeconomic environments. These data allow us to automatically compute each experience’s resonance and representativeness with respect to any expectation, and therefore we can easily scale this exercise: we study 32 experiences, including job loss, sick leave, and loan rejections, across 16 forecasts, including inflation, interest rates, unemployment, and stock returns.

Structurally estimating the model on these SCE data, we find a close correspondence between it and empirical expectations: predicted and actual experience effects are significantly positively correlated for 15 of 16 expectations ($p < 0.01$ for 14, $p < 0.05$ for one). Computing the model’s completeness and restrictiveness, we find that it captures 75.7% of explainable experience-based variation in beliefs while being 99.9% restrictive. The relationship between predicted and actual expectations remains significant even after controlling for time-by-expectation and person-by-expectation fixed effects, showing that it partly reflects within-person changes in cued experiences. These results underscore the portability of the methodology we propose.

This paper contributes to a growing literature studying how cues and associations shape beliefs and choices. A rich body of theoretical work models such effects through

⁴We also ask the LLM to provide brief explanations for its similarity ratings, which we can use to interpret similarity judgments. By automatically extracting features from these explanations and comparing their importance, we find that both humans and LLMs, when making similarity judgments, focus on only a very small number of the many possible features. This connects our approach to a growing literature using LLMs to extract features and contextual structure from complex environments (e.g., Sarkar 2026, Bybee 2025, Bastianello et al. 2025, Charles & Sui 2026).

similarity-based retrieval and simulation (Bordalo et al. 2021, 2023, in press; Kahana 2012; Mullainathan 2002; Bordalo, Burro, Coffman, et al. 2025), and a growing empirical literature documents these forces in the lab (Enke et al. 2024, Graeber et al. 2024, Bohren et al. 2025) and the field (Link et al. 2026, Charles 2025, Colonnelli et al. 2024, Augenblick et al. in press, Ashraf et al. 2025). Economists have begun measuring associations empirically, through similarity judgments (Bordalo, Conlon, et al. 2026, Gennaioli et al. 2024, Bordalo, Burro, Coffman, et al. 2025), shared features (Bohren et al. 2025, Charles & Sui 2026, Bordalo, Burro, Gennaioli, et al. 2025), or LLM embeddings (Sarkar 2026). We show that two theory-derived sufficient statistics—representativeness and resonance—characterize the magnitude of cue-based belief distortions, no matter the complexity of the underlying similarity structure. This insight simplifies the application of these models to both lab and field settings.

More broadly, we contribute to a growing literature on bounded rationality that studies agents who cannot process or take into account all available information when making decisions, for example due to inattention (Caplin et al. 2019, Gabaix 2019, Bohren et al. 2025, Conlon in press, Bordalo, Conlon, et al. 2026), failures of contingent reasoning (Esponda & Vespa 2014, Martínez-Marquina et al. 2019, Niederle & Vespa 2023), cognitive imprecision (Woodford 2020, Enke & Graeber 2023, Oprea 2024, Bordalo et al. in press), directed memory search (Bakhtin et al. 2024), or adopting incomplete mental models (Eliaz & Spiegler 2020, Schwartzstein & Sunderam 2021, Barron & Fries 2025, Charles & Kendall 2025, Augenblick, Backus, et al. 2025). While our focus is on the role of similarity and associations, much of this literature models other factors like salience, endogenous information seeking, or statistical fit in determining what agents consider.

Finally, our work connects to literatures in psychology on anchoring and adjustment (Tversky & Kahneman 1974) and on priming (Meyer & Schvaneveldt 1971). Several leading explanations for the source of anchoring effects argue that anchors bring to mind information consistent with them or possible answers close to them (Wilson et al. 1996, Chapman & Johnson 1999, Mussweiler & Strack 1999). Similar mechanisms are thought to underpin priming effects, whereby the activation of certain words or concepts improves recognition of associated ideas (Collins & Loftus 1975).⁵ Our model provides one possible microfoundation for some of these effects: cues and primes may influence beliefs by shifting which states come to mind. Our methodology may also contribute to a better

⁵Note that such conceptual or semantic priming effects, such as more readily recognizing words semantically related to a prime, are very well-established. So-called behavioral priming effects, where the link between the prime and a behavior of interest is much more tenuous (e.g., between the word “old” and a tendency to walk more slowly out of the lab), are more controversial (see Meyer 2014 for a discussion of these issues).

understanding of the scope of these effects and their magnitude across domains.

2 Model

In this section, we first describe the basic setup of an associative sampling model, which formalizes how an agent constructs beliefs by sampling states of the world. States come to mind more readily when they are more similar to the cues the agent faces, leading to belief distortions. We then derive a simple characterization of belief distortions: two properties of cues—summarizing whether they bring many states to mind and the extent to which those states favor a particular hypothesis—determine both the extent of and the limits to their effects on beliefs. These lead to testable predictions based on the underlying similarity structure of the state space, which can be estimated by eliciting similarity judgments.

2.1 Associative Sampling and Cued Beliefs

Objective environment Let Ω be the set of possible states of the world, with the likelihood (or frequency) of $\omega \in \Omega$ given by $\pi(\omega)$. The agent is forming an expectation over a variable $\psi(\omega)$, whose value is state-dependent. Under the frictionless rational benchmark, the expectation of ψ is then given by $\Psi_{rat} \equiv \sum_{\omega \in \Omega} \pi(\omega)\psi(\omega)$.

Context and cued beliefs In contrast to the rational benchmark, our agent cannot necessarily consider every possibility when forming her beliefs. Instead, she relies on the states that come to mind in the context in which she is forming her beliefs. Following past work (Bordalo et al., 2023, *in press*), we model this as an associative sampling process. We assume the agent has an unbiased “mental database”—she is aware, in a sense, of all possible states ω and their weights $\pi(\omega)$ —but the sampling process that allows her to access these states is distorted by the context C . This assumption is made for parsimony – this implies that the only source of belief distortions comes from the context C , which allows us to focus on the role of the cues. The model can easily be extended to the case where the agent’s mental database can also be influenced by her memories, as is modeled in Bordalo, Burro, Coffman, et al. (2025); Bordalo, Gennaioli, et al. (2026).

The agent samples a state ω from her mental database with probability proportional to its similarity to the context $S(\omega, C)$, which we formalize in greater detail below. In the large sample limit, the agent’s simulated distribution is proportional to:

$$\pi_s(\omega|C) \propto \pi(\omega) \cdot S(\omega, C), \quad (1)$$

We assume that the agent is naive about the context distortion: her beliefs are given by the simulated distribution $\pi_s(\omega|C)$. If sampling is unbiased ($S(\omega, C) = 1$ for all ω), we recover the rational benchmark.

Cues and similarity We model the context C as a set of cues: elements in the agent’s environment that, even when normatively irrelevant, nonetheless can bring certain states to top of mind. For example, these cues might include the agent’s current situation, her recollection of past states, and depictions or descriptions of scenarios (e.g., in advertisements), or experimental stimuli. Formally, let $\tilde{\Omega}$ be the set of all possible cues, where $\Omega \subset \tilde{\Omega}$. This captures two important features of cues. First, known states of the world $\omega \in \Omega$ can act as cues: simply prompting individuals to think about a state they already know to be possible can shift beliefs by ensuring that this state (and ones similar to it) come to mind. Second, however, even cuing an agent with a scenario she knows to be impossible ($\tilde{\omega} \notin \Omega$) can also affect her beliefs, because it can still distort what comes to mind. For example, many advertisements depict scenarios that are far-fetched for typical consumers, or an agent may be recalling a previous experience that she knows cannot exactly happen again, but each of these cues can bring to mind similar possible states.

We take as exogenous a similarity function $S : \Omega \times \tilde{\Omega} \mapsto [0, 1]$, where $S(\omega, \omega) = 1$ for all ω . $S(\omega_1, \omega_2)$ captures the degree to which ω_1 is similar to (or “associated with”) the cue ω_2 . We can then generalize S to apply to subsets of Ω and $\tilde{\Omega}$ by averaging pair-wise similarity. Formally, let A be a subset of Ω and B a subset of $\tilde{\Omega}$. Then, the similarity between A and B is given by:

$$S(A, B) = \frac{1}{|A| \cdot |B|} \sum_{\omega \in A, \tilde{\omega} \in B} S(\omega, \tilde{\omega}). \quad (2)$$

Interpretation The current model naturally has many simplifying assumptions. For example, the similarity function in our model is exogenous, but other work microfound and endogenizes similarity by considering the salience of features that objects have in common (Tversky, 1977; Bordalo, Conlon, et al., 2026). In addition, while for simplicity we describe our agent as forming beliefs about the probability of different states of the world, our framework can be applied more generally. The crucial ingredients are that there be 1) a large set of objects that need to be integrated by the agent, and 2) an *associative structure* between these objects and cues. Though an uncertain state of the world is a natural example where our framework applies, there are others that lack any “true” uncertainty. For example, an agent may be forming beliefs about her own past experiences (Bordalo et al., 2023). Or an agent may be trying to evaluate how often she will

find herself in various situations in the future, about which she may in principle already know enough to decide with certainty. All of the analysis to follow will apply even to such cases where there is no “true” uncertainty.

2.2 Determinants of Cue Distortions

Characterizing cued beliefs Proposition 1 characterizes how the context C distorts an agent’s assessment of $E[\psi(\omega)]$.

Proposition 1. Define $\mathcal{W}(\omega_c) = E[S(\omega, \omega_c)] = \sum_{\omega \in \Omega} \pi(\omega) \cdot S(\omega, \omega_c)$ as the *resonance* of ω_c , and $\mathcal{D}^\psi(\omega_c) = \frac{\text{Cov}[S(\omega, \omega_c), \psi(\omega)]}{E[S(\omega, \omega_c)]}$ as the *representativeness* of ω_c with respect to ψ . The agent’s beliefs $E_s[\psi(\omega)|C]$ in context C are given by:

$$\overbrace{E_s[\psi(\omega)|C] - E[\psi(\omega)]}^{\text{Cue Distortion}} = \frac{\sum_{\omega_c \in C} \mathcal{W}(\omega_c) \cdot \mathcal{D}^\psi(\omega_c)}{\sum_{\omega_c \in C} \mathcal{W}(\omega_c)}. \quad (3)$$

Proposition 1 shows that the effect of $\omega_c \in C$ on the assessment of Ψ is characterized by two summary statistics based on similarity. The first is $\mathcal{W}(\omega_c) = E[S(\omega, \omega_c)]$, which we call the *resonance* of ω_c , the average similarity of ω_c to states in Ω . Because cues bring states to mind in proportion to their similarity, the resonance of ω_c captures the extent to which ω_c brings many states to mind. Importantly, the resonance of a cue is *not its likelihood*: an extreme or even impossible cue ($\pi(\omega_c) = 0$) may still have substantial resonance $\mathcal{W}(\omega_c)$ and sway beliefs if it brings to mind many states. This possibility also emphasizes how cues shape beliefs differently from informative signals: even impossible cues may shift beliefs in predictable ways by shaping what comes to mind.

The second quantity, the cue’s representativeness $\mathcal{D}^\psi(\omega_c)$, captures whether ω_c is disproportionately associated with states with high values of ψ . To build intuition, consider the case in which $\psi(\omega) = I(\omega \in H)$, for $H \subset \Omega$: the agent is assessing the likelihood of a hypothesis H , a subset of the state space. In that case, one can easily show that $\mathcal{D}^\psi(\omega_c) > 0$ if and only if $S(H, \omega_c) > S(\Omega - H, \omega_c)$: the cue ω_c raises the assessment of H if and only if it is more similar to states in H than to states in $\Omega - H$. Thus, if resonance captures the average similarity to ω_c , representativeness captures a notion of *relative* similarity: whether ω_c is disproportionately similar to states in H compared to states outside of H .

Connection to other notions of representativeness By referring to $\mathcal{D}^\psi(\omega_c)$ as representativeness, we draw on Tversky & Kahneman’s seminal formulation that grounded representativeness in terms of similarity. For example, they write that “probabilities are

evaluated by the degree to which A is representative of B , that is, by the degree to which A resembles B ". Because probability assessments have to sum to one, representativeness translates most naturally to a notion of *relative* similarity, which our definition of $\mathcal{D}^\psi(\omega_c)$ captures. Our definition is also tightly connected to the grounding of representativeness in likelihood-ratios, as in [Gennaioli & Shleifer \(2010\)](#) and [Bordalo et al. \(2016, 2018, 2019\)](#): in Appendix B, we show that our definition collapses to (i.e., corresponds one-to-one with) the likelihood ratio in a simplified setting.

Example Equation 3 implies that the effect of context C on the agent's beliefs is a convex combination of the representativeness of each of its constituent cues $\omega_c \in C$, with weights proportional to their resonance. To visualize this, Figure 1 shows an example where an agent is forming beliefs about the probability that the unemployment rate will exceed 5%. Suppose the true distribution is shown in black, such that the rational belief is 50%. The agent's beliefs are distorted by two recent experiences of hers that serve as cues: she recently received a job offer, but she was also denied a loan application.

The red and blue curves in Figure 1 plot the beliefs that each cue would induce absent the other. The job offer brings to mind states with low unemployment, while the loan denial brings to mind states with higher unemployment. The resonance of each cue—how many such states each brings to mind—is given by the total mass under each curve, while the representativeness is given by the ratio between the darker and lighter areas for each curve (minus the truth). As the figure illustrates, the job offer has a less extreme representativeness: her beliefs with only that cue would be closer to the truth than with only the loan denial as a cue. But it also has a higher resonance, which results in beliefs when the context consists of both cues being too low relative to the rational benchmark, despite the loan denial in some sense being a stronger cue.

To summarize, equation 3 provides a simple characterization of how cues affect beliefs. A cue shifts beliefs regarding a variable ψ to the extent it both brings many states to mind and disproportionately brings to mind states with higher (or lower) ψ . The first force is captured by resonance, which determines how much weight a cue receives in the sampling process. The second is representativeness, which captures whether the states evoked by the cue tend to favor one hypothesis over another. Cues that are both highly resonant and highly representative therefore exert the largest effects on beliefs. This example highlights how both objects—representativeness and resonance—are necessary to form predictions about the total distortion an agent's context induces.

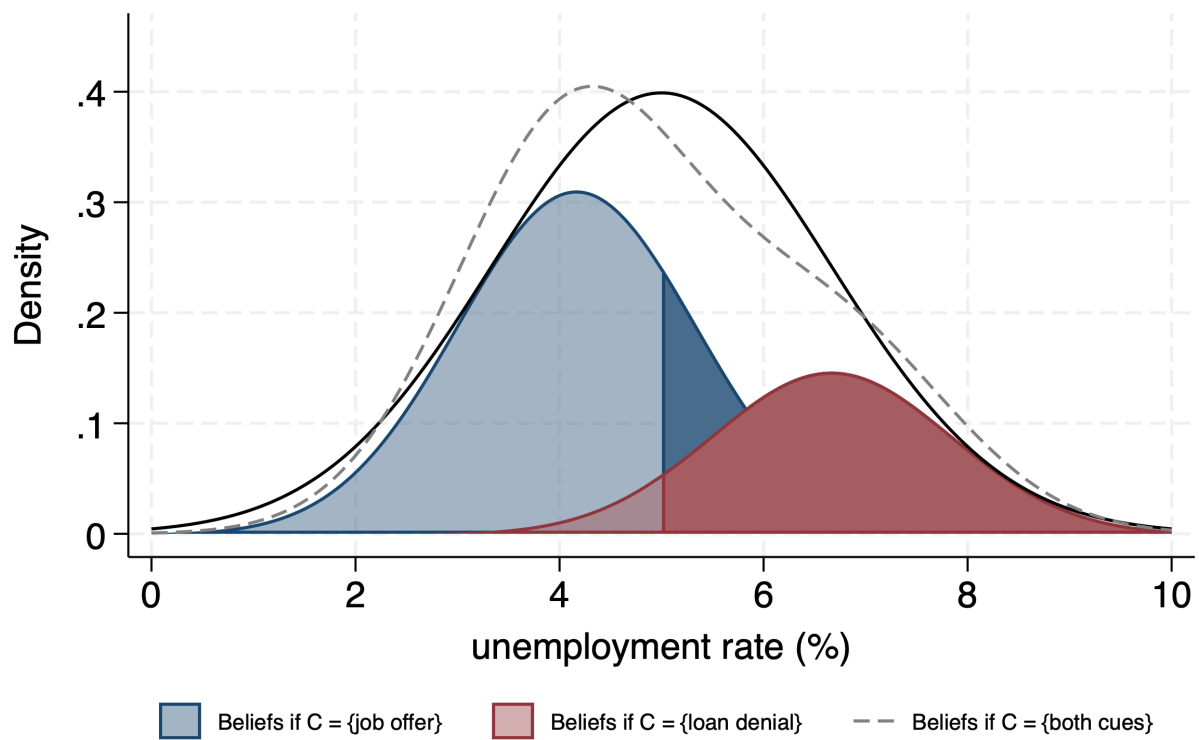


Figure 1: Example: Context-distorted beliefs

2.3 Taking the Model to the Data

The results above characterize how cues distort beliefs depending on their similarity to different states of the world. If similarity and the context were fully observed, computing these predictions would be straightforward. But an agent’s context may contain many unobserved cues and the similarity function linking cues to states could in principle be quite complex. In this section, we discuss how, despite this generality, the right kind of similarity data nonetheless allow us to form testable predictions.

Decomposing the context. Suppose the total context consists of two components, an unobserved latent context $C_0 = \{\omega_1^c, \dots, \omega_N^c\}$ and a marginal cue q , which takes values in $\mathcal{Q}$: $C \equiv C_0 \cup \{q\}$. In this case, Proposition 1 implies the following about how adding q distorts beliefs.

Corollary 1. *The belief distortion from the marginal cue is given by*

$$E_s[\psi(\omega)|C_0 \cup q] - E_s[\psi(\omega)|C_0] = \frac{\mathcal{W}(q)}{\mathcal{W}_{C_0} + \mathcal{W}(q)} (\mathcal{D}^\psi(q) - \mathcal{D}^\psi(C_0)), \quad (4)$$

where

$$\mathcal{W}_{C_0} = \sum_{n=1}^N \mathcal{W}(\omega_n^c) \quad \text{and} \quad \mathcal{D}^\psi(C_0) = \frac{\sum_{n=1}^N \mathcal{W}(\omega_n^c) \mathcal{D}^\psi(\omega_n^c)}{\sum_{n=1}^N \mathcal{W}(\omega_n^c)}. \quad (5)$$

In other words, the entire residual context can be summarized by two parameters: its resonance and its representativeness. The resonance of the latent context is the sum of the resonances of its constituent cues, and the representativeness of C_0 is a convex combination of the representativeness of those cues ω_n^c , weighted by their relative resonance.

Empirical procedure. These observations allow us to derive predictions from the model using the following procedure. First, for each cue $q \in \mathcal{Q}$, we estimate its resonance and representativeness from empirical measurements of similarity. We draw on decades of research in cognitive psychology that has studied associations by analyzing subjective similarity assessments elicited from experimental participants (e.g., [Tversky 1977](#), [Nosofsky 1988](#)), as well as recent work in economics that has begun to adopt these insights ([Bordalo, Conlon, et al. 2026](#); [Bordalo, Burro, Coffman, et al. 2025](#); [Bordalo et al. in press](#)). Concretely, suppose we observe similarity judgments between a cue $q \in \mathcal{Q}$ and a set of states $\omega_1, \dots, \omega_N$ drawn from the true distribution $\pi(\omega)$. These data allow us to estimate

$$\widehat{\mathcal{W}}(q) = \frac{1}{N} \sum_{n=1}^N S(\omega_n, q) \quad \widehat{\mathcal{D}^\psi}(q) = \frac{\sum_{n=1}^N S(\omega_n, q) \psi(\omega_n)}{\sum_{n=1}^N S(\omega_n, q)} - \Psi_{rat}.$$

With these estimates of resonance and representativeness in hand for each $q \in \mathcal{Q}$, one can predict the effect of the entire family of cues on beliefs. This is possible because the model has only two latent parameters, $\mathcal{W}_{C_0}$ and $\mathcal{D}^\psi(C_0)$, even as the number of cues in $\mathcal{Q}$ grows. With data on beliefs in response to many different cues, the model will be overidentified, with many more moments than parameters to be estimated.

3 Experiment

We employ the procedure described above to test the model’s predictions in a controlled environment. Doing so requires several ingredients. First, we need an environment with many possible states—such that participants cannot simply enumerate them all when forming beliefs—but which is simultaneously simple enough to describe that confusion does not drive the results. Second, we need the ability to provide cues that will not alert participants to any states that they are not already aware of or change their beliefs about those states’ likelihood. Finally, we need the ability to elicit similarity judgments between cues and states drawn from the state space. To achieve these goals, we design a novel experimental task, which we describe below.

3.1 Design

The environment The experiment begins by describing to participants a simple data-generating process framed as a “game.” Each game has seven rounds r , in each of which a number of points p_r is added to the “total score” of the game, where $p_r \in \{+1, +2, +3, +4, -5, -6\}$ with each outcome being equally likely and i.i.d. across rounds. To make this process intuitive for participants, it is described as repeatedly rolling a six-sided die. The faces showing one through four are displayed in green, indicating that this number is added to the score, while five and six are shown in red, indicating subtraction. Participants are first tasked with guessing the share of games that end with a total score (adding up across the seven rounds) greater than zero. Afterward, they are also asked what fraction of games end up with an even total score (e.g., -4, -2, 0, 2, 4, ...).

Note that this environment maps directly onto the model described above. In the experiment, the set of states is each of the possible sequences of seven dice rolls: $\Omega = \{+1, +2, +3, +4, -5, -6\}^7$. Further, these states are naturally grouped into focal hypotheses, such as games whose final scores are positive vs negative or are even vs odd. There are also a large number of states— 6^7 , or 279,936, of which 46.2% have a positive final score and 50.0% have an even final score—obviously precluding the strategy of simply enumerating

all states consistent or inconsistent with the hypotheses. If participants form their beliefs by relying on which outcomes come to mind easily, their beliefs may be susceptible to cues.

Cues After learning the rules, all participants are shown a graph to help them visualize the game, as shown in the left panel of Figure 2. Those in the control group, who are shown no additional cue, must wait 20 seconds before a text box appears asking them the average number of games per 100 that end up with positive scores and then a similar question asking about even scores. The remainder of participants are shown a cue before reporting their beliefs. These cues (as depicted in the right panel of Figure 2) are framed as example games that participants watch unfold sequentially. Participants must then wait 20 seconds before reporting their beliefs.⁶

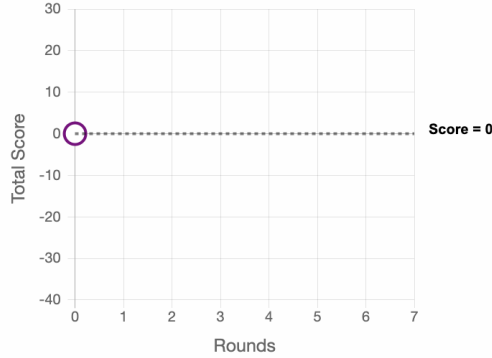
For several reasons, these cues are objectively (and explicitly) uninformative. First, the correct answers to the questions they are answering are (though difficult to ascertain) logically pinned down by the description of the data-generating process, which is fully specified even absent any example. Second, all outcomes are equally likely, so participants cannot infer that some outcomes are more likely from the fact that they were chosen to be shown as examples. And finally, we explicitly (and truthfully) tell all participants that cues are uninformative. They are told it is random whether they will be shown an example and, if so, the type of example that will be shown. Thus there is no objective signal to be extracted from the example game that a participant happens to see. We also explicitly state that “any example(s) you are shown should not impact your guesses,” and participants must confirm their understanding of this fact in a comprehension question before continuing with the experiment. 95.4% of participants get this comprehension question correct on the first try, indicating a very high level of understanding that cues should be irrelevant.

Similarity ratings The effect these cues should have on participants’ beliefs, according to our framework, depends on how similar they are to different possible outcomes. After providing their beliefs, all participants make 15 pairwise similarity judgments between cues and outcomes of the game. Figure A.I shows what these elicitations looked like. Each pair of outcomes that participants compared included one cue from the set of cues shown to participants (described in more detail below) and an example game randomly

⁶Most participants were shown a single cue, but we also included treatments showing participants five and ten of our baseline positive or baseline negative cues (described below). These treatments were designed to test predictions on how cueing effects aggregate when multiple cues are added to agents’ context. We relegate discussion of these treatments to Appendix A.

The chart below may help you visualize how the game works.

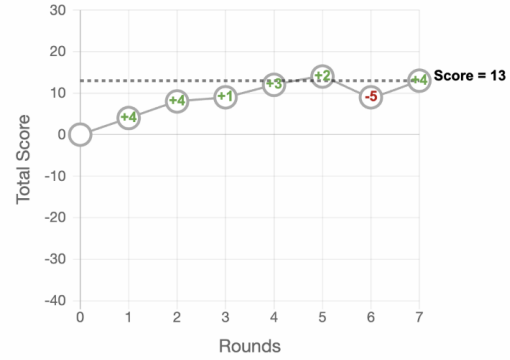
In a moment, we'll ask you some questions about what the final score tends to be. Please take a few seconds to think. The screen will automatically change soon to ask for your answer(s).



For every 100 games, what is your best guess about the number that end with a **final score greater than zero**?

per 100 games

Below you can watch an example game where the final score is 13. Please hover over the purple dots below to start watching it.



In this example game, the **final score was 13**. Please answer the question below.

For every 100 games, what is your best guess about the number that end with a **final score greater than zero**?

per 100 games

Figure 2: Belief elicitation with and without a cue

generated from the true data-generating process. This procedure allows us to generate estimates of the resonance of each cue and its representativeness with respect to any hypothesis, using the Monte Carlo method described in Section 2.3.

Logistics and comprehension A total of 3,151 participants, recruited from Prolific, completed our experiment in March 2025. The median participant took 12.5 minutes to complete the experiment and earned a \$2.00 completion payment. Participants earned an additional \$1.00 bonus if their belief about the fraction of positive or even games (decided randomly) was within 5 percentage points of the truth. Participants had to answer ten comprehension questions correctly before they could continue with the experiment, and were truthfully told that those who gave more than two incorrect answers to comprehension questions would immediately be screened out. This restriction only screened out 3% of participants who attempted to participate in the experiment, indicating a high level of engagement and comprehension. In the main text, we restrict analysis to the 77% of the remaining participants who gave no incorrect answers to any of the ten comprehension questions, though none of our main results are sensitive to this choice.⁷

⁷Screenshots of the experiment, which was conducted through a Qualtrics survey, can be found in [this online document](#). An interactive version of the experiment itself can be accessed (with the ability to choose treatment) by [following this link](#).

3.2 Reduced-Form Evidence

We begin by showing reduced-form evidence using a preregistered set of “baseline” cues. We show that when cues differ greatly in their representativeness, computed using similarity data, they have large differential effects on beliefs. This pattern holds even when these cues are not elements of the state space but rather depict impossible outcomes. In contrast, cues that do not differ in representativeness fail to shift beliefs, even when they are consistent with opposing hypotheses. We then turn to structurally estimating the model.

Baseline cues To construct our baseline cues, we randomly selected ten outcomes each from the true data-generating process from among outcomes with final scores equal to -13, -12, -11, -10, +10, +11, +12, and +13. This process resulted in 80 baseline cues, the final scores of which are half positive, half negative, half even, and half odd. We can then compare the beliefs of the 309 participants assigned to see one of these cues to the 161 participants assigned to the control group (no cue).

The blue dots in Figure 3 split these baseline cues by whether their final score is positive or negative. For each group, the figure shows average resonance (Panel A), representativeness (Panel B), and beliefs about the probability that the final score is positive (Panel C). Three facts stand out. First, positive and negative baseline cues are approximately equally resonant ($p = 0.294$): that is, on average they are equally similar to the true distribution of possible outcomes. Second, however, positive baseline cues are 37.3 p.p. ($p < 0.001$) more representative of outcomes with a positive final score than negative baseline cues are. That is, the positive baseline cues are judged by participants as being more similar to the average outcome with a positive final score than the negative baseline cues are.⁸ Third, Panel C of Figure 3 shows that beliefs differ dramatically across these treatment groups. Participants who see a baseline positive cue believe that 58.9% of games end with a positive score, compared to 39.8% among participants who see a negative cue ($p < 0.001$). Average beliefs in both groups are significantly different from beliefs in the control group (44.5%, different from baseline negative at $p = 0.049$ and from baseline positive at $p < 0.001$).⁹

These effects are striking for two reasons. First, as discussed in greater detail above, these cues are explicitly described to participants as objectively uninformative, and par-

⁸This result is exactly what we would expect if participants tend to judge games as similar to each other when their final scores are close to one another (which we show is largely the case in Section 4).

⁹Note that, because the representativeness of the default context is unobserved, the model does not make unambiguous predictions about what average beliefs in the control group should be.

ticipants had confirmed their understanding of this fact on the page immediately prior to the belief elicitation. Second, these effects are quite large. The main effect of 19.1 p.p. represents 77% of the standard deviation of beliefs in the control group. To provide a benchmark against which to compare this effect size, we can look at updating in response to objectively informative signals from the literature. For example, in Möbius et al. (2022), participants receive a 75%-accurate binary signal about whether their IQ-test score was above the median. Beliefs about the chances of actually having an above-median performance are 48% of a standard deviation higher for those who receive a positive vs a negative signal.¹⁰ Thus, our uninformative cues have substantial effects on beliefs even compared to quite precise objectively informative signals in other contexts.

Impossible cues Recall from Section 2 that a cue need not depict an objectively possible outcome to influence beliefs. Even a cue depicting a state that the agent knows will not occur in the future (e.g., fictional or historical states) can shape beliefs by bringing to mind similar states that are in fact possible. Our experimental paradigm allows us to test this prediction by constructing what we call “impossible” cues. Recall that each round either adds 1, 2, 3, or 4 points, or it subtracts 5 or 6 points. We construct impossible states by altering our baseline cues to explicitly break these rules, adding or subtracting a number of points disallowed by the data-generating process.¹¹ A total of 310 participants are shown one of these impossible cues before stating their beliefs.

Over and above the fact that all cues are already known to be objectively irrelevant, we make the impossibility of these rule-breaking cues extremely salient to participants (see Appendix A for more details), and therefore a natural supposition is that they should not affect beliefs. However, cues depicting impossible outcomes may nonetheless bring to mind possible states similar to them. The red dots in Panel B of Figure 3 show that, indeed, positive impossible cues are more similar to positive possible states than negative impossible cues are: they have a much higher representativeness with respect to positive outcomes than do negative impossible cues ($p < 0.001$). Thus, we should expect positive impossible cues to boost beliefs about the probability of positive outcomes.

This is exactly what we find, as shown by the red dots in Panel C of Figure 3. Participants shown an impossible positive cue believe that the likelihood of a positive score is 54.6%, compared to only 43.8% for participants shown an impossible negative cue

¹⁰This statistic comes from our own calculation using Möbius et al.’s (2022) replication files.

¹¹In particular, we start by randomly picking baseline cues with a final score equal to either -13 or +10. We then randomly choose one round within these cues. If that round added points (+1, +2, +3, +4), we switch it to adding five points (an impossible event given the rules of the game). If that round subtracted points, we switch it to subtracting four points (also impossible given the rules). We randomly generate 80 cues in this way, leading to impossible cues with final scores of -12, -11, -10, -9, +11, +12, +13, and +14.

($p < 0.001$). Thus even explicitly fictional outcomes can shape beliefs by shifting which possible outcomes come to mind.

Ineffective cues We have seen that, when cues differ greatly in their representativeness toward a hypothesis, beliefs shift in tandem. By the same logic, some beliefs should be relatively impervious to cues, if states consistent with a hypothesis are no more similar to each other than they are to those inconsistent with it. In that case, no cues can differentially skew the states that come to mind. To test this, we asked participants their beliefs about the share of games that end with an even vs odd final score. If participants primarily judge the similarity between cues and outcomes according to how close their final scores are (which, as we show in Section 4 below, they largely do), then even and odd games will be very similar to each other. No cue would then be very representative of either group.

Panels A and D of Figure 3 show that, indeed, odd and even baseline cues have very similar resonance ($p = 0.465$) and close to zero representativeness with respect to even outcomes (even and odd cues are depicted with circles and diamonds, respectively).¹² Our model therefore predicts no difference in beliefs about the probability of an even outcome depending on whether participants were shown an even vs odd cue. The blue and red dots in Panel E of Figure 3 show that this is indeed what we find. The average belief about the probability of an even outcome is 45.9% for those shown an odd cue and 46.7% for those shown an even cue ($p = 0.552$). Neither of these averages is significantly different from beliefs in the control group (44.9%, $p = 0.315$ and $p = 0.558$ for even and odd, respectively). Thus, not only do cues move beliefs when the similarity data suggest they *ought* to (positive vs negative cues), they largely fail to move beliefs when the similarity data say they ought *not* to (even vs odd cues).¹³

¹²Even cues do have a significantly higher representativeness with respect to even outcomes than do odd cues ($p = 0.002$), but this difference is very small: 1.1 percentage points, 97% smaller than the difference in representativeness between positive and negative baseline cues. Note that, because beliefs are a convex combination of the representativeness of cues, our model therefore predicts at most a 1.1 percentage point difference in beliefs across these cues, much smaller than what we are powered to detect with our baseline cues (MDE = 4.3 p.p.).

¹³We show in Table A.I that in the expanded sample of cues that we use to structurally estimate the model, we do find a small (about 3 p.p.) but significant effect of possible even cues on beliefs about the likelihood of even outcomes. We do not find such an effect for the impossible cues, however, and though insignificant the point estimate is in the opposite direction.

3.3 Estimating the Model

The results above show reduced-form evidence largely in line with the model described in Section 2. But how well quantitatively does our model fit the data? To answer this question, we structurally estimate the model on an expanded set of cues. In addition to the 160 baseline possible and impossible cues described in the previous section, we collected data on beliefs in response to 200 other cues, chosen partly at random.¹⁴ In total, we estimate the model using data from 1,862 participants answering two different questions (about the likelihood of positive vs negative and even vs odd outcomes) facing 360 distinct cues, plus the control group who saw no additional cue. We estimate the parameters of the model therefore to match 722 moments in the data. Our model, in contrast, has only three degrees of freedom, each pertaining to the unobserved default context: its resonance, representativeness with respect to positive vs negative outcomes, and representativeness with respect to even vs odd outcomes.

The other inputs to the model—the resonance and representativeness of each of the cues—are pinned down by the similarity judgments that we elicit (i.e., separately from the beliefs data). Panel A of Figure 3 shows—in addition to the baseline cues depicted in blue and red—the estimated resonance of all other cues in gray. We see that outcomes closer to “the middle,” with final scores near zero, have greater resonance. In Panel B of Figure 3, we see that representativeness with respect to positive outcomes is monotonically increasing in the final score of cues. This result is intuitive: as a cue’s final score becomes more positive, the “nearby” states tend to be positive, leading to a higher representativeness.

In contrast, Panel D of Figure 3 shows that, unlike with positive vs negative outcomes, even and odd baseline cues tend to have approximately zero representativeness with respect to even vs odd outcomes. In fact, only 6% of cues have a representativeness toward even outcomes that is statistically distinguishable (at $p < 0.05$) from zero, indicating that this distribution is quite similar to what we would expect if they all were in fact zero.¹⁵

¹⁴In particular, for each possible final score (besides those already included in our baseline cues) we randomly selected two cues from the set of possible outcomes with that score. There were two exceptions to this procedure. First, we instead sampled 40 cues each (instead of only two) with final scores of -1 and +1 to have more precise data on cues close to the boundary of our main beliefs question (positive vs negative outcomes). Second, there is only one possible cue each for the most extreme outcomes (final scores of -42 and +28). Because we also wanted more precision at these extremes, participants were more likely to be assigned to these extreme cues than to any other particular cue. Excluding these extreme cues, on average 4 participants were exposed to each cue, whereas 162 and 154 participants were assigned to the extreme negative and positive cues, respectively.

¹⁵For comparison, 86% of cues have a representativeness toward positive outcomes that is statistically significantly different from zero. That said, we only barely fail to reject ($p = 0.053$) that the representativeness of all cues with respect to even outcomes is jointly zero.

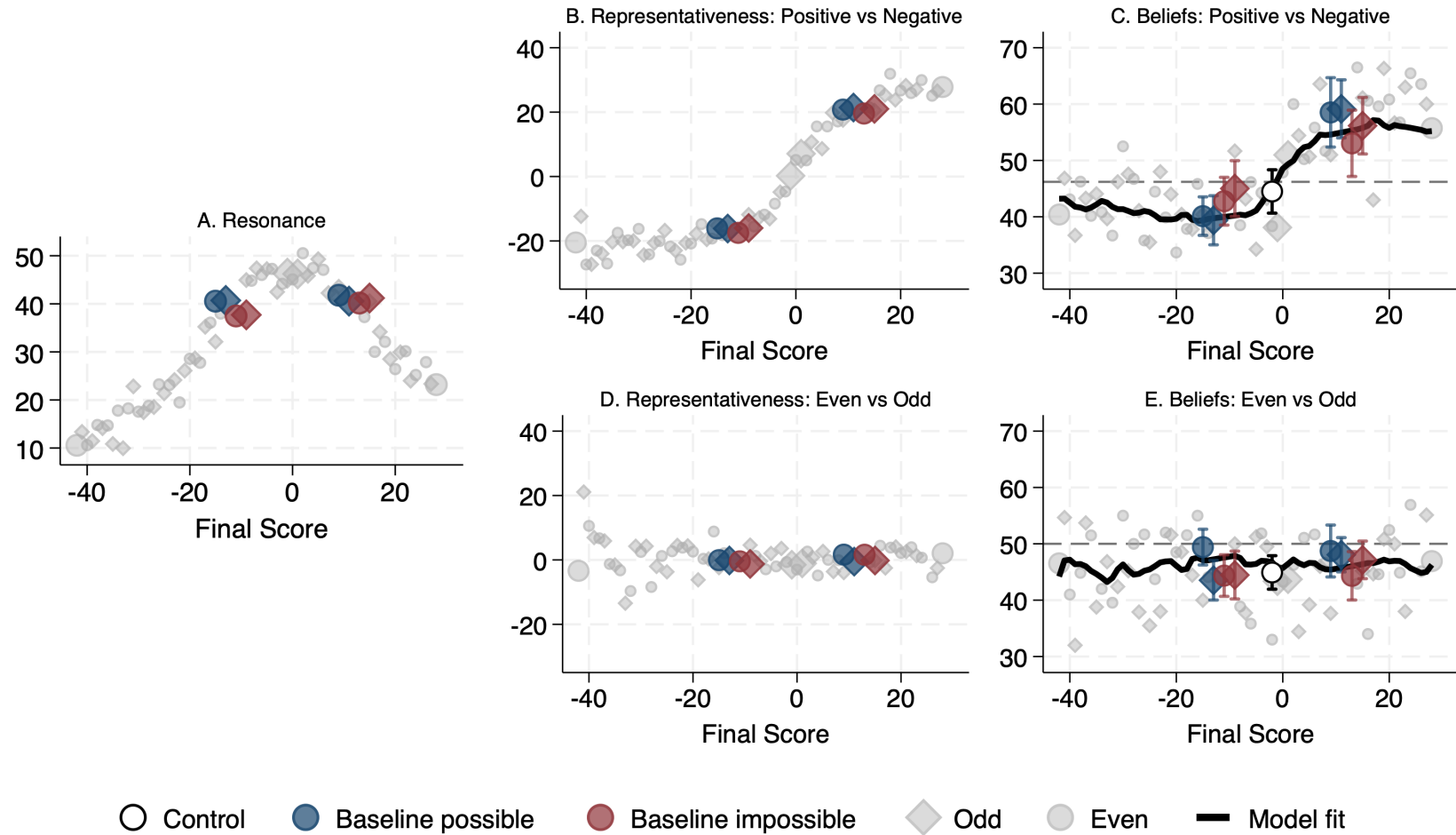
All three of these results make sense given that participants in large part judge similarity by looking at the final score of outcomes (see Section 4 below).

Parameter estimates and fitted vs empirical beliefs We estimate the model’s three free parameters by minimizing the squared distance between predicted and empirical beliefs, and we construct confidence intervals by bootstrapping the data. We estimate that the representativeness of the default context (i.e., absent additional cues) toward positive and even outcomes is 0.006 (95% CI = [-0.011, 0.022]) and -0.063 (95% CI = [-0.076, -0.052]), respectively. These estimates, as expected, closely match the average (lack of) bias in the control group for the positive vs negative and even vs odd beliefs. The resonance of the default, which governs how effective cues are in changing beliefs, we estimate at 51.8% (95% CI = [40.8%, 65.6%]), slightly larger than that for our baseline cues (see Figure 3).

The black curve in Panel C of Figure 3 shows that the model’s predictions for beliefs about positive vs negative outcomes closely track average beliefs in the data. For intermediate cues, the model correctly predicts that beliefs about positive outcomes rise steeply with the final score: average beliefs climb from 39.8% for baseline negative cues to 51.0% for cues with a final score of +1 ($p < 0.001$ vs. baseline negative) to 58.9% for baseline positive cues ($p = 0.003$ vs. +1; $p < 0.001$ vs. baseline negative). For more extreme cues, however, the model predicts a tradeoff between representativeness and resonance: despite their more extreme representativeness, these cues should have no larger, indeed slightly smaller, effects, because their resonance is much lower. Again, this is what we find. Our most extreme cues (final scores of -42 and +28) have directionally (though insignificantly) smaller effects than the baseline cues, and in both cases we can reject that the gap in beliefs is as large as the gap in representativeness ($p = 0.024$ for negative cues, $p < 0.001$ for positive cues). This result confirms that representativeness matters less when resonance is low.

In contrast to beliefs about positive vs negative outcomes, Panel E of Figure 3 shows that beliefs are largely the same across even vs odd cues (depicted in circles and diamonds, respectively). The model successfully predicts this null effect, which follows from even and odd cues each having approximately zero representativeness with respect to even outcomes.

Figure 3: Resonance, Representativeness, and Beliefs Across Cues



Notes: This figure shows average resonance of all cues (Panel A), their representativeness with respect to positive outcomes (Panel B), their representativeness with respect to even outcomes (Panel D), average beliefs about the probability of positive outcomes in response to each cue (Panel C), and average beliefs about the probability of even outcomes in response to each cue (Panel E). The curves in Panels C and E show the fitted model predictions from our structural estimation. For visual simplicity, we show averages across cues with the same final score, with the size of dots corresponding to the sample sizes. This figure drops a single outlier (cues with a final score of 25) from Panel C, where the average belief (among six participants) was 75.8%.

Completeness and restrictiveness of the model How successful is the model in explaining the data? Following [Fudenberg et al. \(2022\)](#), we compute its completeness and restrictiveness. The completeness of model $\mathcal{M}$ is given by how much $\mathcal{M}$ improves the expected loss relative to a baseline model $\mathcal{M}_0$ (which in our case we take to be the rational benchmark), as a fraction of the improvement given by a “best-performing” model $\mathcal{M}^*$ (which we estimate by random forest). We also compute the restrictiveness of the model, how poorly it fits synthetically generated fictitious data.¹⁶

We estimate the expected loss of each model $\mathcal{M}_0, \mathcal{M}, \mathcal{M}^*$ based on a 10-fold cross-validation. We estimate that our model is 81.8% complete (95% CI = [71.2%, 87.7%]) and 97.6% restrictive (95% CI = [97.4%, 98.1%]). That is, it achieves 81.8% of the additional explanatory power (over the Bayesian benchmark) of the machine-learning benchmark despite many fewer degrees of freedom. For comparison, [Fudenberg et al. \(2022\)](#) study the completeness of various classic behavioral theories: they estimate that gambler’s/hot-hand fallacy is 7-10% complete in explaining participants’ generation of “random” outcomes, that Poisson cognitive hierarchy models (PCHMs) are 68-97% complete in explaining initial play in games, and that cumulative prospect theory (CPT) is 94% complete in predicting certainty equivalents. Our model thus achieves completeness (in our novel domain) within the range of these more established theories. But this high level of completeness is not a result of the model being capable of explaining any data: our restrictiveness estimate of 97.6% indicates that the model only improves on the Bayesian benchmark by 2.4% in our synthetic data. For comparison, [Fudenberg et al. \(2026\)](#) find that CPT is only 27% restrictive, while PCHMs are about 97% restrictive.

3.4 Other Outcomes

Process data: what came to mind? In our model, states similar to the cues the agent faces come to mind more readily when she is forming beliefs. To provide suggestive evidence pointing toward this mechanism, we asked participants, on a separate page after they had answered both beliefs questions, to indicate an outcome that “you felt came to mind for you” while forming their beliefs. Figure A.VIII shows that treatment effects on these data strongly track effects on overall beliefs. Participants shown a positive baseline

¹⁶Formally, let E_M be the expected prediction loss (according to the true DGP $\pi(D)$) of model $\mathcal{M}$: $E_M \equiv E_\pi[\mathcal{L}(D|\mathcal{M})]$. The completeness of model $\mathcal{M}$, $\kappa(\mathcal{M})$ is given by $\kappa(\mathcal{M}) = \frac{E_{\mathcal{M}_0} - E_M}{E_{\mathcal{M}_0} - E_{\mathcal{M}^*}} \in [0, 1]$. For restrictiveness, we follow [Fudenberg et al. \(2026\)](#) and estimate the restrictiveness of the model by calculating its completeness with respect to synthetic data. In particular, we create a “jumbled” dataset D^o , where we match each cue with the resonance and representativeness estimates of a randomly selected other cue. Let E_M^o be the expected loss on the jumbled dataset. Then, our restrictiveness measure becomes: $\rho^M \equiv \frac{E_M^o}{E_{\mathcal{M}_0}^o}$.

cue are 31.6 p.p. more likely ($p < 0.001$) to indicate that a positive outcome came to mind than those shown a negative baseline cue. In contrast, those shown an even baseline cue are not more likely to report that an even outcome came to mind than those shown an odd cue (difference = 3.6 p.p., $p = 0.376$). See Appendix A for more details.

Confidence We also investigate the correlation between beliefs and subjective uncertainty (Enke & Graeber, 2023). We asked participants how certain they were that their answer to each question was correct. We find no effect of providing a cue on participants' confidence: those in the control group are 13% of a standard deviation *more* confident in their beliefs than those who receive a cue, though this difference is not statistically significant ($p = 0.124$). Relatedly, one might wonder whether our null effects of cues on beliefs about even vs odd outcomes are driven by participants' being more confident *ex ante* about this question than about positive vs negative outcomes. Though we do find somewhat higher average confidence for the even vs odd question than the positive vs negative question (56.1 vs 52.4 on a 0-100 scale, or 14% of a standard deviation), Figure A.IV shows that there is no relationship between confidence and treatment effects on beliefs. Taken together, we interpret these results as suggesting little link between confidence and the effects of our cues on beliefs.

3.5 Discussion: Alternative Explanations

Standard updating Our design rules out rational Bayesian updating as driving any of our results. In particular, there is no objective uncertainty to begin with, as the right answers to all questions are logically pinned down by the description of the data-generating process. Our cues furthermore represent states that a rational agent already knows are possible (or not, in the case of our impossible cues). There is thus no signal for a rational agent to react to.

Non-standard updating Given that people are not perfectly rational Bayesians, a natural question is whether our results might instead reflect a boundedly rational updating procedure. One interpretation of associative sampling is that it is one form of bounded rationality: our agent cannot consider every possible state, and so she forms beliefs based on those possibilities that come to mind. This might represent a reasonable heuristic to solve a difficult problem given cognitive constraints.

Our results do provide evidence against certain other non-standard learning stories, however. For example, they are unlikely to represent simple confusion or misinterpretation of our cues as being objectively informative signals. Participants are told cues should

not affect their beliefs, and they confirm their understanding of this fact with a comprehension question immediately prior to forming their beliefs. Furthermore, we restrict our main analysis to the large majority of participants who answer every comprehension question, including this one, correctly on the first try. Lastly, the effect of cues varies systematically across our treatments, which is inconsistent with a mechanical updating heuristic, such as updating by a constant amount.

One may still wonder if a different model of quasi-Bayesian updating could provide a more natural interpretation of our findings. We briefly describe the main patterns of effects (and null results) that such an alternative would need to predict. Some beliefs react strongly to objectively uninformative cues (Panel C of Figure 3) while certain other beliefs do not (Panel E of Figure 3). Furthermore, these effects on beliefs occur without systematic effects on certainty (Figure A.IV), as most natural stories of updating from signals would predict.¹⁷ Effects on beliefs are also mirrored in the process data we collect: i.e., participants *say* that positive outcomes come to mind after being presented with positive cues (Figure A.VIII). Finally, the size of these effects quantitatively matches our theory quite closely—following from independent similarity measurements—even when considering a wide range of cues. While with sufficient flexibility it might be possible to describe an alternative theory accounting for these patterns, we show that they follow naturally from a standard model of associative sampling.

Noise and inattention Next, our results are also inconsistent with being driven by noise or inattention. In addition to the fact that we focus on participants who score perfectly on all comprehension questions (and thus to that extent cannot be described as inattentive), noisy or inattentive data are typically thought to attenuate the responsiveness of decisions relative to an objective benchmark (e.g., Gupta et al. 2023; Enke et al. in press). In contrast, we often find *overreaction* to treatment variation, because the objective benchmark is that beliefs should be insensitive to cueing. This result is reminiscent of Augenblick, Lazarus, & Thaler (2025), but recall that despite all our cues being uninformative, we only sometimes find overreaction to cues: participants exhibit no sensitivity to even vs odd cues.

¹⁷In particular, if participants react to cues as though they are informative signals, we would naturally expect participants in the control group (who received no additional “signal”) to exhibit lower confidence in their beliefs than those in the treatment group. This follows from the law of total variance if we assume that our cues were drawn randomly from the set of possible “signals.” Of course, making this assumption requires committing a category error, as our cues are not in fact signals at all.

Experimenter demand Finally, while there is increasing evidence that experimenter demand effects are not a salient concern in economics lab experiments (de Quidt et al. 2018, Winichakul et al. 2024), several facts additionally speak against this force driving our results. First, we explicitly tell participants that cues should not affect their beliefs, so if anything experimenter demand may push against finding any effects of cues. This is particularly true in the case of impossible cues, where we stress that the cues depict outcomes that are not possible given the rules of the game. Second, if experimenter demand were driving our results, we might expect all cues (both even vs odd and positive vs negative) to have similarly sized effects. Instead, however, only some cues are effective at changing beliefs: in particular, these tend to be the ones the theory predicts should matter more.

4 LLM Similarity

The previous section showed that similarity judgments, when combined with the model from Section 2, largely succeed in predicting the effects of cues on beliefs. A natural worry, however, is that collecting similarity data will be costly or infeasible in some settings. In this section, we therefore explore whether large language models (LLMs) can substitute for human similarity judgments. This investigation follows a growing literature using LLMs to extract structured features and contextual information in field settings (e.g., Sarkar 2026, Bybee 2025, Bastianello et al. 2025).

Eliciting LLM Similarity. We elicit similarity from an LLM (Claude Haiku 4.5) using a prompt meant to replicate as far as possible the human similarity elicitation (see Figure A.V for the exact prompt). We elicit 36,270 ratings (the same number as with our human data), each time between a randomly selected cue and a random outcome from the true data-generating process. We also include images of the cue and outcome in the prompt by encoding each pair as a side-by-side two-panel chart depicting the cumulative score path, with round-by-round changes annotated in green (gains) or red (losses) and a dashed line marking the final score. An example of such an image is shown in Figure A.VI.

Predicting cue effects using LLM similarity. This procedure yields 36,270 pairwise similarity judgments between cues and states. We can now ask whether these data succeed in predicting the effect of cues on human beliefs. Apart from replacing human similarity judgments with LLM-generated ones, the remainder of the analysis proceeds exactly as in Section 3. In particular, we use these similarity ratings to estimate the res-

onance and representativeness of each cue and feed these quantities into the structural model to generate predicted beliefs.

Figure A.II summarizes the results. Panels A, B, and D compare the resonance and representativeness of cues estimated using LLM similarity judgments with those estimated from human data. Panels C and E plot predicted beliefs implied by the model using LLM similarity in blue and human similarity in black. Despite relying entirely on LLM-generated similarity judgments, the resulting predictions track the empirical patterns in the beliefs data quite well. Relative to the model using human similarity judgments (a completeness of 81.8% and restrictiveness of 97.6%), we find that the model estimated with only LLM similarity achieves a completeness of 75.0% and a restrictiveness also of 97.6%. Thus, while LLM similarity is slightly less predictive than human judgments, it nonetheless captures most of the explanatory power of the model.

Which features drive similarity judgments? We can also use our LLM data to interpret the similarity judgments we collect. In addition to eliciting a numeric similarity rating, we also ask the LLM for a short explanation of its answer. In Appendix A.3, we describe a procedure for automatically extracting features of cues and outcomes from these explanations. Note that the set of possible features in our setting—functions from sequences of cumulative scores across rounds to real numbers—is very large, and our procedure uncovers 208 unique features mentioned by the LLM. These include the final score of the game, its variance, its largest decrease, and many more.

Of course, not all of these features are equally important in explaining similarity judgments. To explore their relative importance, we conduct a principal component analysis (PCA). In particular, for each of our 208 features f_k , we compute the difference between each cue i and state j from the similarity ratings (pooling human and LLM judgments): $|f_k(i) - f_k(j)|$. Table 1 shows the ten principal components that explain the largest share of the variance in feature differences along with two individual feature differences they are most correlated with. For example, the first two components mostly load on differences in final score and on the share of rounds in which the score is positive.

To quantify the relative importance of different features, we then compute Shapley values for each principal component (Shorrocks, 1982): how much additional variation in similarity judgments can be explained by adding a component to a regression that already includes a random subset of others.¹⁸ Figure A.III shows each principal compo-

¹⁸More precisely, we regress similarity judgments on all 2^{10} subsets of the first ten principal components. For each of these components, we then calculate the average increase in R^2 from adding it to the set of regressors. We then divide this average increase by the overall R^2 of the regression that includes all ten components. This yields a measure of the fraction of explainable variation in similarity judgments that can

PC	Top 2 Correlated $ \Delta\text{Feature} $ (correlation)		Var. Explained
1	Final Score (+0.95)	Total Losses (+0.93)	31.4%
2	Share Positive (+0.79)	Positive Minus Negative Count (+0.78)	9.3%
3	Largest Decrease (+0.74)	Monotone Increasing (+0.72)	8.2%
4	Score–Round Correlation (+0.64)	Position of Minimum (+0.58)	6.0%
5	Late Minus Early Change (+0.47)	Early Minus Late Avg. Change (+0.44)	2.7%
6	Late Minus Early Change (+0.50)	Midpoint Minus Final (+0.48)	2.4%
7	First-Round Sign (+0.45)	Any of Rounds 1–3 Positive (+0.45)	2.0%
8	Mid-Game Max Minus Early Min (+0.40)	Change in Avg. Slope (+0.39)	1.9%
9	Inverse Avg. Absolute Change (+0.54)	Sum of Absolute Changes (+0.40)	1.9%
10	Sign Changes (incl. zero) (+0.55)	Sign Changes (excl. zero) (+0.49)	1.6%

Table 1: Top 10 PCA Feature Vectors

nent’s Shapley value for explaining both human (gray bars) and LLM (blue bars) similarity data. Two results stand out. First, both human and LLM similarity judgments respond strongly to the first principal component, which loads largely on differences in the final score between the cue and the example (see Table 1). This commonality between humans and LLMs explains the broadly aligned patterns of resonance and representativeness in Figure A.II. Second, there are subtle differences between human and LLM similarity. For example, human similarity data appear more “sparse” than LLM similarity: a very large majority (74%) of the variation in human judgments can be explained by the first principal component, while only 47% of LLM judgments are explained by this component.

5 Field Application: Macro Expectations

The previous sections studied a controlled lab-experimental environment. We now ask whether the same framework can be applied in the field to study how personal economic experiences shape macroeconomic beliefs (Malmendier & Nagel, 2011; Kuchler & Zafar, 2019; ?; Cenzone, 2026; Bordalo, Gennaioli, et al., 2026). A central question in the experience-effects literature revolves around scope: which experiences affect which beliefs? Our framework provides a method for predicting the answer to this question “from scratch” using similarity data, without relying on *ex ante* intuitions to identify likely hypotheses. By using LLMs to generate the similarity data, we can also automate this process and therefore deploy it at scale: generating and testing predictions about the effects of many different experiences on many different types of expectations.

be attributed to each component.

Data. We use the New York Fed’s Survey of Consumer Expectations (SCE), a monthly rotating panel spanning 2013–2024 that elicits households’ macroeconomic expectations as well as information about their recent economic experiences. We construct 32 binary experience indicators. These include job loss, maxing out a credit card, being in poor health, missing a rent payment, going on sick leave, and many more. We then analyze respondents’ answers to 16 macroeconomic forecast questions, including about inflation, interest rates, the unemployment rate, and ease of credit access (see Tables D.I and D.II for the full list of experiences and expectations). We pair each forecast outcome with a corresponding historical macroeconomic series spanning 1961–2023 (e.g., CPI, federal funds rates, S&P 500 returns). We standardize participants’ beliefs about each macroeconomic series to have a mean of zero and a standard deviation of one.

Predicting experience-based cue effects. What effect should each of these experiences have on respondents’ expectations? Treating each experience j as a distinct possible cue, we should expect effects on outcome o according to Equation (6):

$$\overbrace{E_s[\psi_o | J_i] - E[\psi_o]}^{\text{Experience effects from cues } J_i} = \frac{W_C \cdot \mathcal{D}^{\psi_o}(C) + \sum_{j \in J_i} \mathcal{W}(c_j) \cdot \mathcal{D}^{\psi_o}(c_j)}{W_C + \sum_{j \in J_i} \mathcal{W}(c_j)}, \quad (6)$$

where C is the context absent any of these cues, and W_C and $\mathcal{D}^{\psi_o}(C)$ are its resonance and representativeness for outcome o . Similarly, $\mathcal{W}(c_j)$ and $\mathcal{D}^{\psi_o}(c_j)$ are the resonance and representativeness of cue j .

Similarity measurement. Computing these objects requires similarity judgments between each cue and the relevant state space. We define the “state space” Ω as each year between 1961 and 2023 along with the value of the historical macroeconomic series during that year. We then elicit similarity judgments between each state in Ω and each experience. In particular, we describe to an LLM the SCE question and response that defines the experience, along with a list of the macroeconomic indicators for the target year, and ask for a similarity rating between 0 and 100. Because many fewer ratings are necessary than for our controlled experiment (63 years $\times$ 32 experiences = 2,016 pairs to be rated, compared to 36,270 ratings before), we elicit similarity ratings from each of seven LLM raters: Claude Haiku 4.5, Claude Sonnet 4.6, GPT-5, GPT-4.1, Gemini 3.1 Pro, Gemini 2.5 Pro, and Gemini 2.5 Flash.¹⁹ We compute the resonance and representativeness for each experience-outcome pair directly using these similarity ratings. We then fit the

¹⁹Our results look very similar if we restrict to only Claude Haiku 4.5, the LLM we used for the experiment. See Appendix C.3 for the exact prompt we use and other details regarding the field exercise.

17 parameters of the model—the resonance of the default context and its representativeness for each outcome—by minimizing the squared difference between model-implied and empirical beliefs, across all 2,305,018 respondent $\times$ survey $\times$ belief observations.

Predicted vs observed experience effects. Figure 4 plots, for each of our 16 forecast variables and 32 experiences, the model-predicted difference in average beliefs for those with and without that experience on the x-axis along with the actual difference in average beliefs on the y-axis. We see that predicted experience effects positively and significantly predict actual experience effects for 15 of our 16 outcomes (14 at $p < 0.01$, one at $p < 0.05$).²⁰ We estimate a completeness of 75.7%, indicating that the model captures the majority of predictive power of respondents’ experiences in explaining their expectations, and a restrictiveness of 99.9%, confirming that the model is not simply flexible enough to explain any mapping between similarity judgments and experience effects.

Many of these predicted experience effects are clearly domain specific: for example, respondents who recently lost their jobs expect the unemployment rate to increase, and those who are late repaying a loan expect credit conditions to worsen. But others are consistent with affect mattering over and above the specific content of experiences or outcomes. For example, those on sick leave expect prices (medical and otherwise) to increase and for credit access to worsen, and those who experienced a credit application denial expect the stock market to decline.²¹

Table 2 regresses respondents’ beliefs on the model-implied predictions. Column 1 shows an estimated slope of 1.07 ($p < 0.001$) after controlling for date-by-outcome fixed effects. These controls confirm that it is within-period differences in experiences (rather than experiences being correlated over time with aggregate changes in expectations) driving the results shown in Figure 4. Column 2 further adds person-by-outcome fixed effects, where we still see a positive and significant coefficient of 0.21 ($p < 0.001$). This coefficient exploits only variation within person in experiences, indicating that a given person’s beliefs move as their reported experiences change month to month.

Taken together, these results show in a field environment that generating similarity data provides a simple method of predicting cue-based effects. We are of course not the first to document experience effects (Malmendier & Nagel, 2011; Kuchler & Zafar, 2019; Taubinsky et al., 2026; Cenyon, 2026; Bordalo, Gennaioli, et al., 2026). But the simplicity

²⁰The only exception is home prices. This appears to be due to the fact that many negative experiences are judged as being similar to the Great Recession, when home prices were largely falling, and yet empirically respondents with negative experiences tend to expect higher home price growth.

²¹Note that some of our experiences are plausibly endogenous to respondents’ macro expectations (e.g., purchasing a home, or searching for a job). Excluding these leaves all results essentially unchanged.

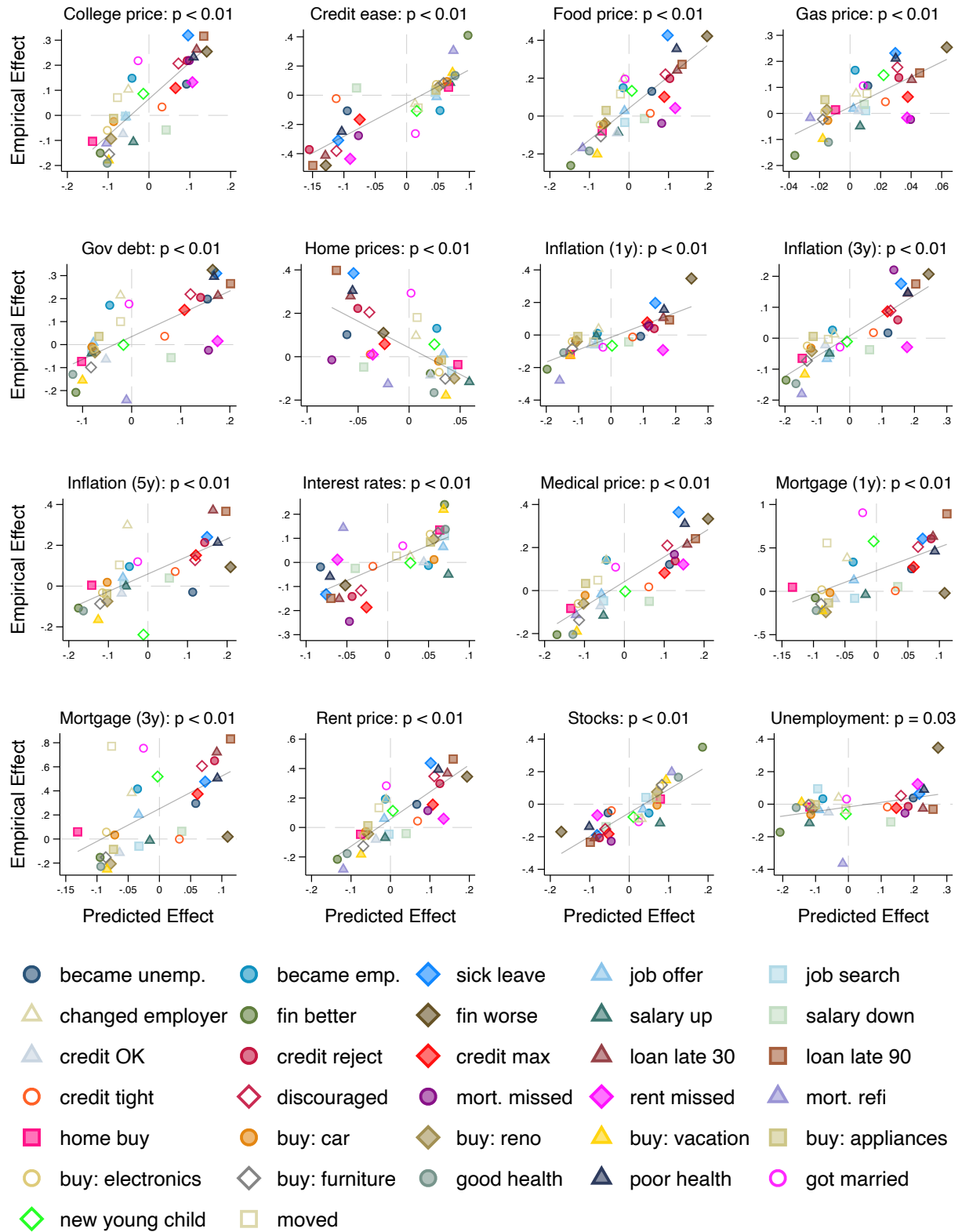


Figure 4: Predicted vs observed experience effects, by outcome. Each panel represents a macroeconomic expectation, and each dot represents a personal experience. The y-axis shows the difference in expectations between SCE respondents who report the experience and those who do not, while the x-axis shows the model's predicted difference from Equation (6).

	(1)	(2)
Predicted belief	1.072*** (0.143)	0.208*** (0.029)
Date $\times$ outcome FE	Yes	Yes
Person $\times$ outcome FE	No	Yes
Observations	2,305,018	2,265,616
Individuals	24,091	20,184
Outcomes	16	16

Table 2: SCE beliefs regressed on the structural-model prediction. Outcome and predictor are standardized within outcome. The predictor is $\widehat{\text{belief}}_{io}^z$ from Equation (6), evaluated at the estimated $\widehat{W}_C$ and $\widehat{D}^{\psi_o}(C)$. SEs are two-way clustered by individual and by outcome. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

of our method—and the fact that it can be applied without taking any *ex ante* stand on which experiences are likely to matter for which expectations—allows us to investigate these forces more broadly as well as predict their scope and magnitude.

6 Conclusion

In this paper, we explore how to bring theories of cue-based beliefs to the data. Drawing on a standard model of associative sampling, we show that two sufficient statistics—resonance and representativeness—characterize belief distortions and can be estimated from simple elicitations of similarity judgments. This provides a tractable method of deriving testable predictions from the theory. We showcase this method in a controlled experimental setting, where we find that these predictions are largely confirmed both qualitatively and quantitatively. We also show that LLM similarity judgments help not only to explain our lab experimental evidence but also to predict, in the field, a large number of experience effects on macroeconomic expectations.

Two directions for future work appear especially promising. First, in this paper we focus on an exogenously given and fixed associative structure between states. This is obviously a simplification, as the perceived similarity between objects is a function of the features agents happen to be attending to (Tversky 1977). Introducing endogenous attention, and therefore similarity, would enrich the model: cues might not only shape which states initially come to mind, but also which features agents attend to while evaluating their options, which in turn may shape the associations between different states (see Bordalo et al. in press for related ideas). Second, our work raises the possibility of predicting

treatment effects of cues in a wider range of applied settings, such as in financial investment or advertising. Measuring and deploying similarity judgments to map out these associations and the corresponding effects of cues “in the wild” is an open and exciting avenue for future research.

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A Additional Experimental Details and Results

A.1 Additional Tables and Figures

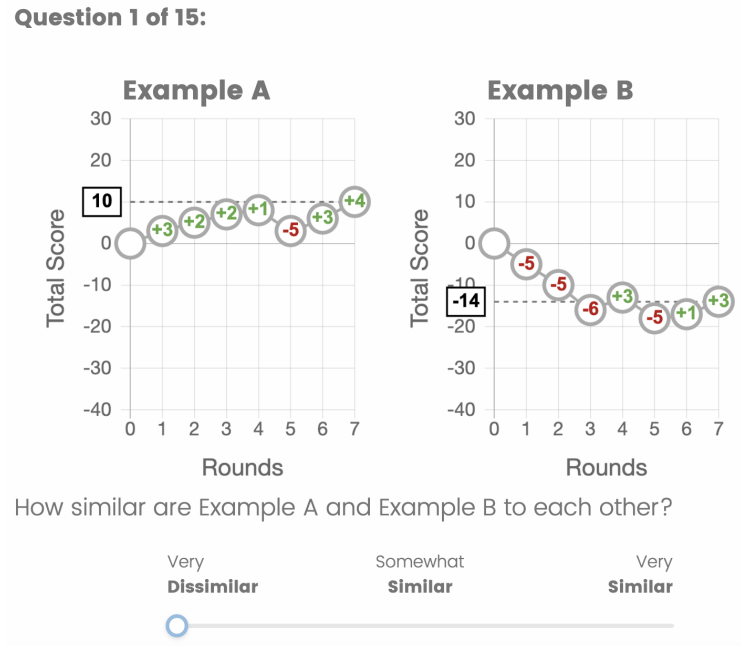
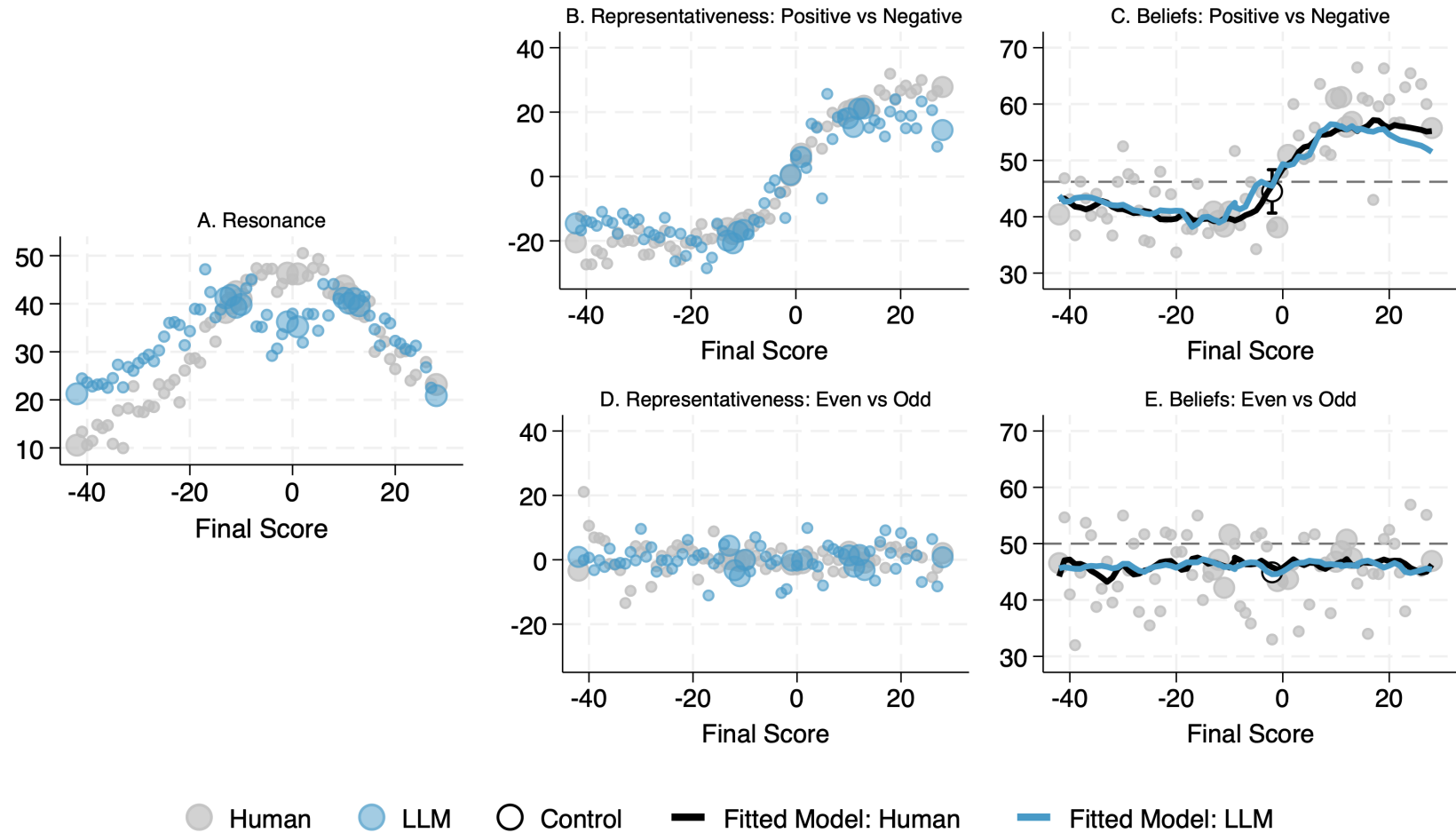


Figure A.I: Example of a Similarity Elicitation

Figure A.II: Human vs LLM Similarity: Resonance, Representativeness, and Beliefs



Notes: This figure overlays the similarity measures computed from our human similarity elicitations (gray) and from LLM similarity elicitations (blue), in the same format as Figure 3. Panels A, B, and D show resonance and representativeness; Panels C and E show average beliefs, with fitted model predictions estimated separately from the human (black curve) and LLM (blue curve) similarity data. As in the main figure, dots average across cues with the same final score, sized by sample size, and the open marker is the control group.

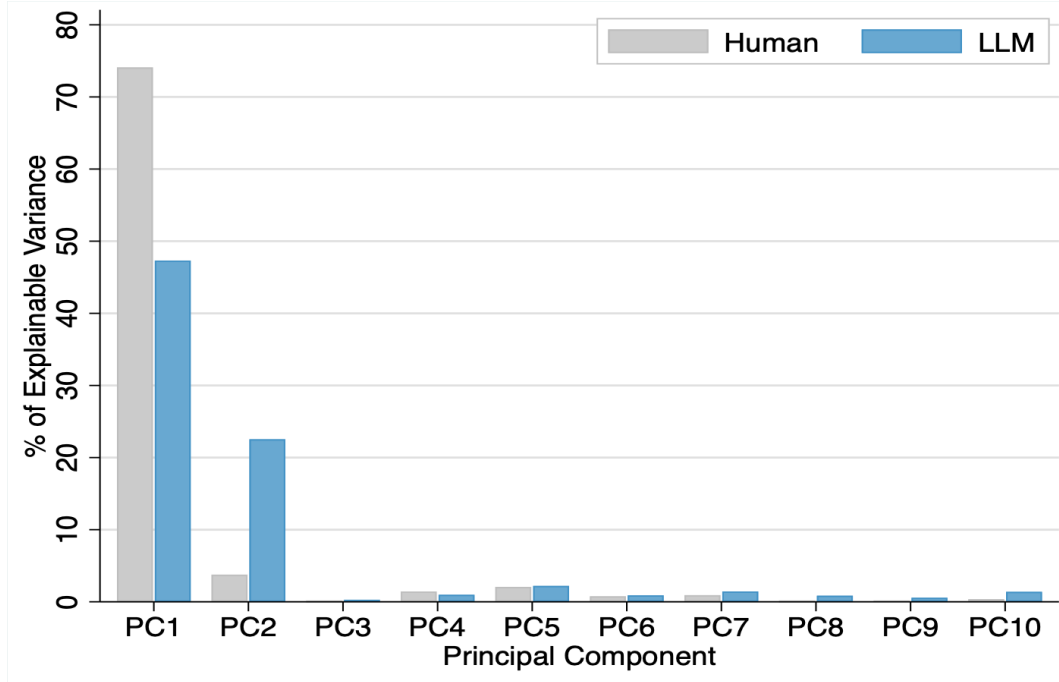


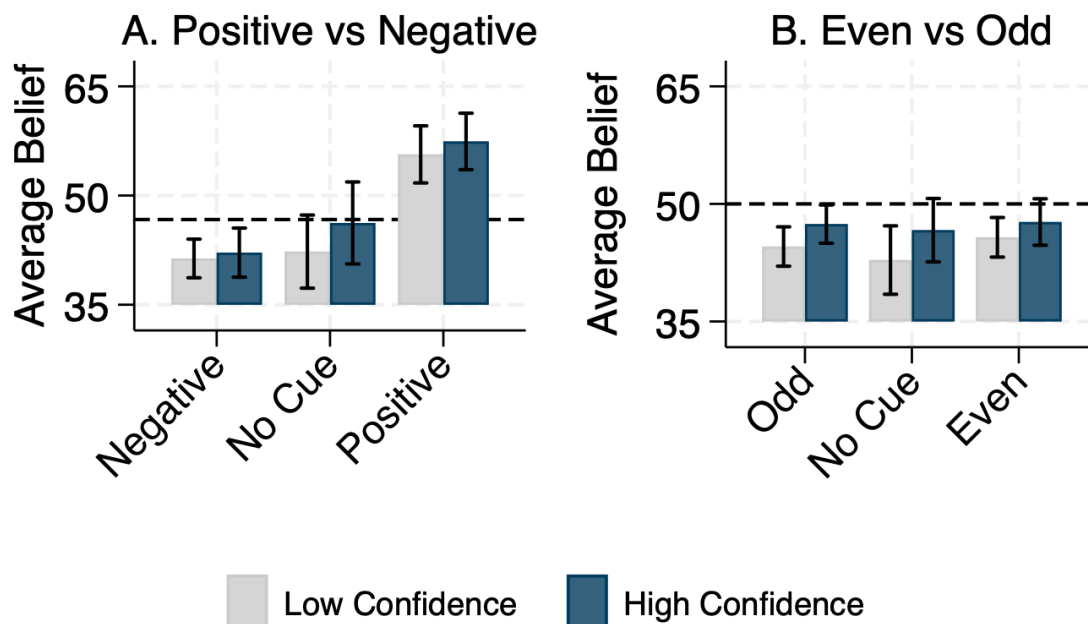
Figure A.III: Feature Importance for Similarity Measures

A.2 Supplemental Analysis and Experimental Details

Impossible cues. In Section 3.2, we mentioned that the fact that our impossible cues were disallowed by the rules of the game was made extremely salient to participants. The full survey instrument can be accessed at [this online document](#), but here we provide more details about how this was done. The text preceding an impossible cue reads, “You’ll notice that **this example game will not follow the rules perfectly**: one of the rounds **adds 5** [or **subtracts 4**] points, which is not one of the possible die rolls.” We then include a comprehension question where participants must correctly report during which round an illegal move was made.²² During the waiting period between revealing the cue and eliciting beliefs, the text reads “In this example game, the final score was X, but remember it **does not perfectly follow the rules**.” Then, right above the question eliciting their beliefs, the experiment tells participants to “focus again **only on games that actually follow all the rules**” (all bolded text is as it appears in the experiment).

²²87% answer this question correctly on the first attempt, and results are unchanged if we focus only on this group.

Figure A.IV: Confidence Does Not Predict Treatment Effects



Notes: This figure shows average beliefs split by whether participants reported low or high confidence in their beliefs (below vs above the median). Panel A shows beliefs about the probability of a positive outcome for negative, no-cue (control), and positive baseline cues, pooling possible and impossible cues; Panel B shows beliefs about the probability of an even outcome for odd, no-cue, and even baseline cues. Whiskers are 95% confidence intervals and the dashed lines mark the objective probabilities.

You will be answering questions about an abstract "game" as part of a research study. There are seven "rounds" in this game. In each round a randomly chosen number of points is either added or subtracted from the total score of the game. Each round, the computer rolls a fair six-sided die, meaning each of its six sides is equally likely to be rolled. On the die the numbers 1 through 4 appear in green, while 5 and 6 are in red. Whenever a green number is rolled, that number of points gets added to the game's "score". Whenever a red number is rolled, that number is subtracted from the game's score. So, while each round is somewhat more likely to gain rather than lose points, losses are somewhat larger than gains. The score starts at zero, so it can end up either positive or negative. Human participants in this study were asked to guess what share of games have a final score (adding up across all seven rounds) that is positive vs negative and that is even vs odd.

You'll be shown pairs of example games and asked to rate how similar or dissimilar they are to each other on a scale from 0 (dissimilar) to 100 (similar). Use a wide range of numbers, and avoid latching onto round numbers.

There are no right or wrong answers for these questions. Please just answer in a way that makes sense to you.

Look at the two example games shown in the image (Example A on the left, Example B on the right).

In Example A, the score at the end of each round was: ...

In Example B, the score at the end of each round was: ...

Please rate how similar these two games are to each other on a scale from 0 to 100.

Return EXACTLY two lines in this format (no extra text): Rating: <integer 0 to 100>

Reason: <one sentence (max 25 words) explaining the rating>

Figure A.V: Prompt for Eliciting LLM Similarity

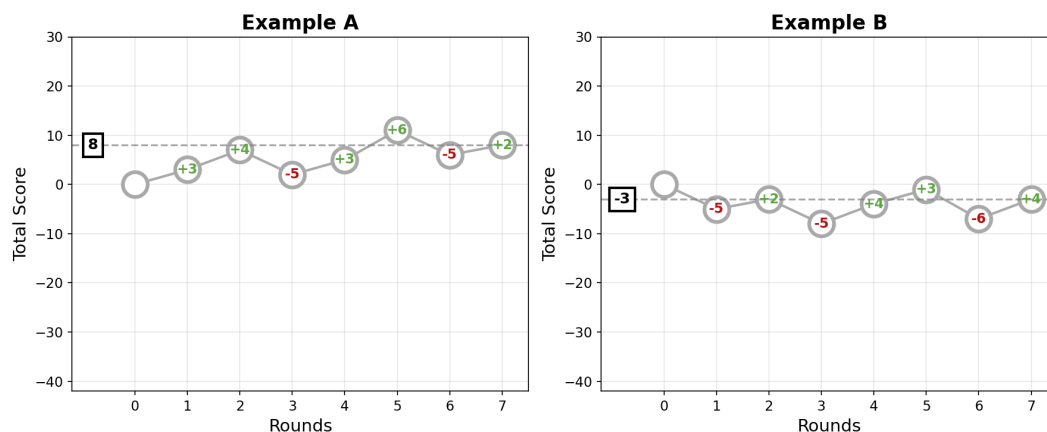


Figure A.VI: Example pair of games as shown to the LLM. Each panel displays the cumulative score over seven rounds, with round-by-round changes annotated in green (gains) or red (losses).

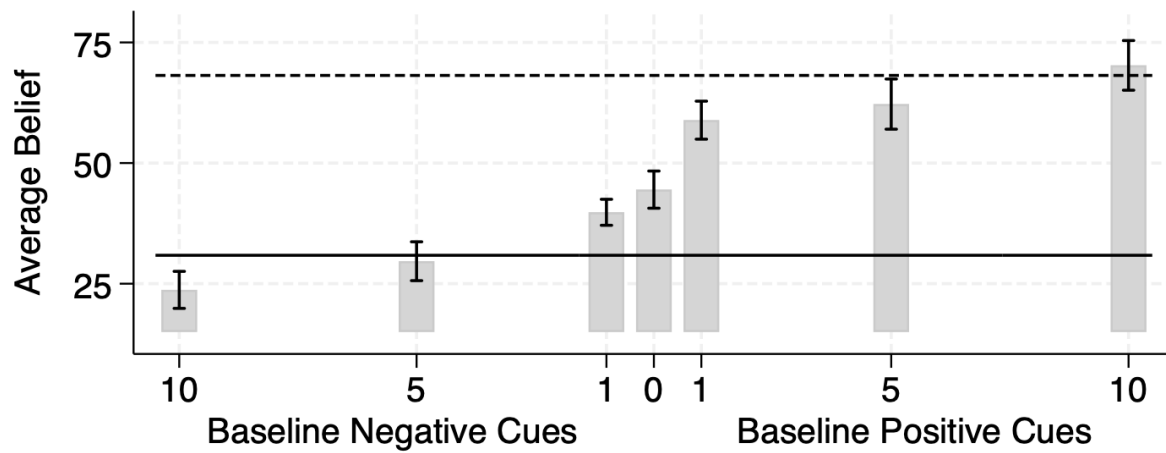
Multiple cues. What effect should providing multiple additional cues have on beliefs? Belief distortions are a convex combination of cues' representativeness, and therefore are bounded by the smallest and largest representativeness among cues in the context, as shown in equation 7:

$$\min_{\omega_c \in C} \mathcal{D}^\psi(\omega_c) \leq E_s[\psi(\omega)|C] - \Psi_{rat} \leq \max_{\omega_c \in C} \mathcal{D}^\psi(\omega_c). \quad (7)$$

Recall that some participants were shown either one, five, or ten baseline cues (either all positive or all negative). These cues were chosen randomly either from the set of positive baseline cues or the set of negative baseline cues. Each cue a participant sees therefore in expectation has the same representativeness as any others they were shown. We can thus test whether belief distortions continue to move toward this representativeness as the number of cues increases, and the rate at which they do so. Equation 7 indicates that we should expect additional cues to further shift belief distortions toward (but not substantially past) the representativeness of cues.

Figure A.VII shows average beliefs across these treatments. Focusing first on the positive baseline cues, we see that average beliefs increase slightly from 58.9% for a single cue to 62.2% for five cues, though this difference is not statistically significant ($p = 0.316$). With ten cues, they further increase to 70.2% (different from a single cue at $p < 0.001$ and from five cues at $p = 0.032$), close to the implied maximal distortion of 67.9% ($p = 0.372$). For negative cues, moving from one to five negative cues reduces beliefs from 39.8% to 29.7% ($p < 0.001$), approximately reaching the maximal representativeness-implied distortion, which for the baseline negative cues is 30.7%. Moving to ten cues reduces beliefs by a further 5.9 percentage points ($p = 0.037$) and therefore has a slightly larger effect than our model predicts.

Figure A.VII: The Effect of Multiple Cues on Beliefs about $P(\text{Final Score} > 0)$

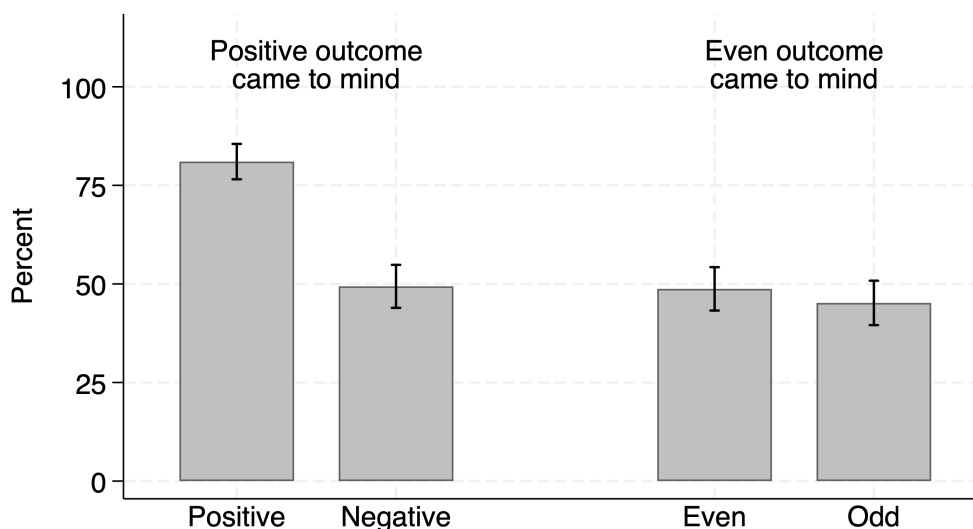


Notes: This figure shows the maximal representativeness-implied distortion from our baseline positive and negative cues (dashed and solid lines, respectively), and average beliefs about the probability of positive outcomes in response to zero, one, five, and ten negative and positive baseline cues.

Process data. In Section 3.2, we described data on which outcomes participants felt “came to mind” when they were forming their beliefs. We elicited these data after participants reported their beliefs. They used an interactive graph to “draw” an outcome of the game by choosing the number of points added/subtracted in each round. We can thus record whether the outcome they indicate is positive vs negative and is even vs odd, and how this tendency differs by the type of cue they were earlier shown.

Note that, while these data are consistent with the mechanism we propose, these process data are not conclusive. In particular, there is no objective benchmark to compare these data to, and so it is unclear how participants “ought” to answer this question. Eliciting which unrealized possibilities come to mind differs in this respect from, say, free-recall measures in related models of memory retrieval (e.g., [Bordalo et al. 2023](#), [Conlon 2025](#)), where incentivizable measures of both beliefs and recall can be independently elicited and compared. Nonetheless, we view our data on which outcomes come to mind as providing suggestive evidence consistent with the model we describe.

Figure A.VIII: Process Data: Which Outcomes Came to Mind?



Notes: This figure shows the percent of participants who, when asked to indicate an outcome that came to mind when forming beliefs, indicated an outcome with a positive final score (left bars) or with an even final score (right bars), by the sign and parity of the cue they were shown, pooling baseline possible and impossible cues. Whiskers are 95% confidence intervals.

Comparing model-predicted and empirical beliefs. Table A.I shows OLS regressions where the dependent variable is participants' beliefs (about positive outcomes in columns 1-4, even outcomes in columns 5-6, and combining data on both in columns 7-8) and where the independent variables include the fitted model's predictions of those beliefs. We see in column 7 that, combining the two belief questions we elicit, the model's predictions on average fit the data: for every percentage point increase in beliefs that the model expects, actual beliefs on average increase by 1.00 percentage points. As expected, this fit is largely driven by beliefs about positive outcomes (which the model can explain through differences in representativeness, see column 1) rather than by beliefs about even outcomes (where, given that representativeness toward even outcomes tends to be zero, the model largely expects no differences in beliefs across cues, see column 5).

Furthermore, note that the predictive power of the model appears robust to including proxies for several other natural heuristics that could drive beliefs. For example, columns 2 and 8 show that the model's predictions remain significant even after including dummies for whether the cue shows an outcome that is "consistent" with the hypothesis being asked about (note that the definition of consistency is not especially clear for impossible cues, which are strictly speaking inconsistent with every hypothesis). Similarly, beliefs about positive outcomes do not seem well described by a simple anchoring-and-adjustment story, where participants' beliefs react only to the final score of the cue they are shown (columns 3 and 4).

Table A.I: Predicted vs Empirical Beliefs

	P(Final Score > 0)				P(Final Score is Even)		Combined	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Model Prediction	1.05*** (0.07)	0.43** (0.18)	1.05*** (0.12)	0.43** (0.18)	0.08 (0.20)	0.14 (0.23)	1.00*** (0.07)	0.80*** (0.07)
Possible × Consistent		10.89*** (2.30)		11.00*** (2.51)		3.18*** (0.84)		4.90*** (0.75)
Impossible × “Consistent”		3.99 (3.75)		46.50 (36.94)		-1.62 (1.88)		-2.05 (1.78)
Possible × Final Score			0.05 (0.03)	-0.00 (0.03)				
Impossible × Final Score			-0.26* (0.13)	-1.79 (1.54)				
Possible		-4.62*** (1.62)	-1.61 (1.34)	15.61 (17.95)		-1.70 (1.39)		-3.74*** (1.16)
Control (No Cue)		2.22** (1.00)	-2.51*** (0.61)	2.29* (1.23)		1.08 (0.88)		1.62*** (0.50)
Constant	-2.41 (3.50)	26.81*** (7.09)	-0.63 (6.06)	6.35 (19.47)	42.11*** (9.33)	39.23*** (10.67)	0.15 (3.06)	10.45*** (3.38)
Observations	1,862	1,862	1,862	1,862	1,862	1,862	3,724	3,724
R ²	0.08	0.10	0.09	0.10	0.00	0.01	0.05	0.06

Notes: This table shows OLS regression estimates. The dependent variable is participants’ beliefs about the probability that the game’s final score is positive (columns 1-4), even (columns 5-6), and both (columns 7-8). “Model Prediction” is the fitted structural model’s prediction for each belief. “Possible” is an indicator for whether the cue a participant was shown was a possible cue. “Consistent” is an indicator for whether the cue was consistent with the hypothesis the participant was considering (i.e., a positive or even outcome). Note that this variable name appears in quotation marks when applied to an impossible cue, as it is unclear whether to consider, for example, a positive-but-impossible outcome consistent with the hypothesis the participant is considering (which concerns only possible outcomes). “Final Score” is the final score of the cue. “Control” is an indicator for whether participants were shown zero cues (the control group). For such participants, we impute values of zero for all other variables in the regression. Only participants who saw zero or one cue are included in these regressions. *, **, and *** indicate statistical significance at the $p < 0.10$, 0.05 , and 0.01 levels, respectively.

A.3 Extracting Features

Here we describe our procedure for extracting features to interpret our similarity judgments. Our prompt eliciting similarity from the LLM also asks the model for a short explanation of its rating. We use these explanations to automatically identify features that contribute to these similarity judgments. We do this by feeding these explanations to another LLM that is asked to identify features that the explanation appears to invoke. Here, features are functions from sequences of cumulative scores in each round to real numbers, and the LLM specifies each feature by writing Python code that computes its value given any sequence of seven scores. See Figure A.IX for the exact prompt.

We extract features in this way from a random 1,000 of our LLM similarity judgments. See Figure A.X for examples of the features extracted from three explanations. This process produces 4,000 features, though these contain many duplicate features because the explanations often overlap. For example, the LLM determines that 98.5% of explanations mention the final score of the examples as a relevant feature. Furthermore, some features that are not exactly identical are nonetheless collinear: for example, one identified feature computes the overall trend (final score minus the first round) while another computes the average trend (which equals the overall trend divided by six). We de-duplicate the list of features like these by dropping any feature that is collinear—across all 279,936 possible games—with an existing feature. This yields 208 unique features.

You are analyzing a similarity judgment that was made by another Claude instance.

This is data from a study where Claude rated the similarity of pairs of “game” sequences.

The game works as follows: There are 7 rounds. Each round, a 6-sided die is rolled. Sides 1-4 (green) add that many points. Sides 5-6 (red) subtract that many points. The score starts at 0.

Each sequence (Example A and Example B) is a list of 7 cumulative scores after each round. For instance, [3, 6, 4, 8, 2, 5, 9] means: after round 1 the score was 3, after round 2 it was 6, etc.

Claude was shown these pairs as graphs and asked to rate similarity from 0 (dissimilar) to 100 (similar).

Here is one pair that was rated:

Example A: {values_a}
Example B: {values_b}

Similarity rating given: {rating}/100
Reasoning provided: “{reason}”

Based on this rating and reasoning, identify up to 4 “features” that appear to be driving the similarity judgment.

IMPORTANT: Each feature must be a function that takes a SINGLE sequence and outputs a value for that sequence. We will later compute similarity by comparing feature values across two sequences.

A feature is a function that takes ONE example vector and outputs either:

- A real number (e.g., final score, volatility)
- A 0 or 1 indicator (e.g., whether it ends positive, whether the final value is even)

Output your response as a JSON object with this exact structure:

```
{
  "features": [
    {
      "name": "short_snake_case_name",
      "description": "Brief description of what this
                    feature captures",
      "code": "def feature_name(seq):\n    return ..."
    }
  ]
}
```

Requirements for the code:

- Each function must take exactly ONE argument: seq (a list of 7 integers representing cumulative scores)
- Each function must return a single number (int or float)
- Use only standard Python (no imports needed beyond math basics)
- Keep functions simple and self-contained
- The function name in the code should match the “name” field

Examples of valid features:

- Final score: `def final_score(seq): return seq[-1]`
- Ends positive: `def ends_positive(seq): return 1 if seq[-1] > 0 else 0`
- Final value is even: `def final_is_even(seq): return 1 if seq[-1] % 2 == 0 else 0`

Focus on features that are ACTUALLY suggested by the reasoning provided. Don’t invent features that aren’t supported by the text.

Respond with ONLY the JSON object, no other text.

Figure A.IX: Prompt for Extracting Features from LLM Explanations

Figure A.X: Examples of Feature Extraction from LLM Explanations

Explanation 1.

Rating: 42/100

Example A: [-6, -12, -18, -24, -30, -36, -42]

Example B: [-6, -4, -9, -14, -20, -25, -23]

LLM explanation: *“Both games start identically but diverge significantly; Example A shows consistent -6 losses while Example B has variable outcomes and ends higher.”*

Extracted features:

```
1. def final_score(seq): return seq[-1]
2. def first_round_score(seq): return seq[0]
3. def consistency_of_changes(seq): changes = [seq[i] - seq[i-1] for i in range(1, len(seq))];
   variance = sum((c - sum(changes)/len(changes))**2 for c in changes) / len(changes); return
   variance
4. def trend_direction(seq): changes = [seq[i] - seq[i-1] for i in range(1, len(seq))];
   all_same_sign = all(c < 0 for c in changes) or all(c > 0 for c in changes); return 1 if
   all_same_sign else 0
```

Explanation 2.

Rating: 28/100

Example A: [2, 4, 7, 8, 11, 5, 8]

Example B: [2, -4, -2, 0, 3, -3, -8]

LLM explanation: *“Both start identically (+2) but diverge sharply; Example A remains consistently positive while Example B turns negative, showing fundamentally different trajectories.”*

Extracted features:

```
1. def first_round_score(seq): return seq[0]
2. def final_score(seq): return seq[-1]
3. def proportion_positive_rounds(seq): return sum(1 for score in seq if score > 0) / len(seq)
4. def score_volatility(seq): import math; if len(seq) < 2:; return 0; changes = [seq[i] -
   seq[i-1] for i in range(1, len(seq))]; mean_change = sum(changes) / len(changes); variance
   = sum((c - mean_change) ** 2 for c in changes) / len(changes); return math.sqrt(variance)
```

Explanation 3.

Rating: 34/100

Example A: [3, 5, 7, 8, 3, 6, 11]

Example B: [2, 3, 5, 8, 10, 12, 14]

LLM explanation: *“Both end positive but differ significantly in volatility, trajectory smoothness, and mid-game performance; Example A has a major setback while B steadily climbs.”*

Extracted features:

```
1. def final_score(seq): return seq[-1]
2. def volatility(seq): return sum(abs(seq[i] - seq[i-1]) for i in range(1, len(seq)))
3. def trajectory_smoothness(seq): reversals = 0; for i in range(1, len(seq) - 1):; if (seq[i] -
   seq[i-1]) * (seq[i+1] - seq[i]) < 0:; reversals += 1; return -reversals
4. def major_setback_magnitude(seq): return min(seq[i] - seq[i-1] for i in range(1, len(seq)))
```


B Proofs

B.1 Section 2 Proofs

Proof of Proposition 1 The agent's assessment of $E[\psi]$ is given by:

$$\begin{aligned} E_s[\psi(\omega)|C] &= \frac{\sum_{\omega} \pi(\omega) \cdot S(\omega, C) \cdot \psi(\omega)}{\sum_{\omega} \pi(\omega) \cdot S(\omega, C)} = \frac{E_{\pi}[S(\omega, C) \cdot \psi(\omega)]}{E_{\pi}[S(\omega, C)]} \\ &= \frac{\text{Cov}_{\pi}[S(\omega, C), \psi(\omega)] + E_{\pi}[S(\omega, C)]E_{\pi}[\psi(\omega)]}{E_{\pi}[S(\omega, C)]} \\ &= \frac{\text{Cov}_{\pi}[S(\omega, C), \psi(\omega)]}{E_{\pi}[S(\omega, C)]} + E[\psi(\omega)], \end{aligned} \quad (8)$$

where E_{π} refers to the expectation taken over ω with measure $\pi(\omega)$. Thus, we obtain:

$$E_s[\psi(\omega)|C] - E[\psi(\omega)] = \frac{\text{Cov}_{\pi}[S(\omega, C), \psi(\omega)]}{E_{\pi}[S(\omega, C)]}. \quad (9)$$

To conclude, recall $S(\omega, C) \propto \sum_{\omega_c \in C} w(\omega_c) \cdot S(\omega, \omega_c)$, which implies $E_{\pi}[S(\omega, C)] = \mathcal{W}(C) = \sum_{\omega_c \in C} w(\omega_c) \cdot \mathcal{W}(\omega_c)$, and by the linearity of the covariance function, one obtains:

$$E_s[\psi(\omega)|C] - E[\psi(\omega)] = \frac{\sum_{\omega_c \in C} w(\omega_c) \cdot \text{Cov}_{\pi}[S(\omega, \omega_c), \psi(\omega)]}{\sum_{\omega_c \in C} w(\omega_c) \cdot \mathcal{W}(\omega_c)} = \frac{\sum_{\omega_c \in C} w(\omega_c) \cdot \mathcal{W}(\omega_c) \cdot \mathcal{D}^{\psi}(\omega_c)}{\sum_{\omega_c \in C} w(\omega_c) \cdot \mathcal{W}(\omega_c)}, \quad (10)$$

as desired. For the special case $\psi(\omega) = I(\omega \in H)$, we have:

$$\mathcal{D}^{\psi}(\omega_c) = \frac{\sum_{\omega \in H} S(\omega, \omega_c) \pi(\omega)}{\mathcal{W}(\omega_c)} - \pi(H). \quad (11)$$

B.2 Connections between notions of representativeness

Consider the case in which an agent is forming beliefs regarding the likelihood of a hypothesis H , $\psi(\omega) = \mathbb{I}(\omega \in H)$, and is cued by a trait T , which we model as $\mathcal{T} \subset \Omega$, the set of states possessing the trait T . $\mathcal{D}^H(\mathcal{T})$ then takes the form

$$D^H(\mathcal{T}) \equiv \frac{\pi(H)(1 - \pi(H))}{\mathcal{W}(\mathcal{T})} \cdot \underbrace{\left[S(H, \mathcal{T}) - S(-H, \mathcal{T}) \right]}_{\text{Is } T \text{ more similar to } H \text{ or } -H?}. \quad (12)$$

Equation (12) formalizes [Kahneman & Tversky \(1972\)](#)'s notion of representativeness as a judgment of *relative* similarity: a trait is representative of H to the extent that it is more

similar to states in H than to states in $-H$.

Imposing more structure on the similarity function links $\mathcal{D}^H(\mathcal{T})$ directly to diagnostic expectations (Bordalo et al., 2018, 2019). Formally, let $h(\omega) \equiv I(\omega \in H)$ and $t(\omega) \equiv I(\omega \in \mathcal{T})$. Suppose similarity between states takes the form $S(\omega, \omega') = \delta_H^{|h(\omega) - h(\omega')|} \delta_T^{|t(\omega) - t(\omega')|}$, where parameters δ_H and δ_T govern how sharply the hypothesis and the trait discriminate across states. For simplicity, suppose $\pi(H) = \pi(-H) = \frac{1}{2}$, and $P(T | H) = q$ and $P(T | -H) = 1 - q$, so that the likelihood-ratio is $\mathcal{R}(T | H) = \frac{q}{1-q}$. Representativeness then takes the form

$$\mathcal{D}^H(\mathcal{T}) = \frac{1}{2} \cdot \frac{\mathcal{R}_S - 1}{\mathcal{R}_S + 1}, \text{ where } \mathcal{R}_S = \frac{q + (1-q)\delta_T}{(1-q) + q\delta_T} \cdot \frac{q + (1-q)\delta_H}{(1-q) + q\delta_H}. \quad (13)$$

The expression in (13) nests diagnostic expectations as a special case. When both the hypothesis and the trait are maximally distinctive ($\delta_H = \delta_T = 0$), the similarity-augmented ratio collapses to $\mathcal{R}_S = \mathcal{R}(T | H)^2$, and

$$\mathcal{D}^H(\mathcal{T}) = \frac{1}{2} \frac{\mathcal{R}(T | H)^2 - 1}{\mathcal{R}(T | H)^2 + 1}. \quad (14)$$

In this limit, representativeness is a function of the likelihood ratio alone: $\mathcal{D}^H(\mathcal{T})$ is positive, implying overreaction, if and only if $\mathcal{R}(T | H)$ exceeds one, and increases monotonically with the likelihood ratio, recovering the intuition of diagnostic expectations. More generally, however, our notion of representativeness depends on similarity in a way that the likelihood ratio does not capture on its own. Even holding the likelihood ratio fixed, $\mathcal{D}^H(\mathcal{T})$ is decreasing in both δ_T and δ_H : overreaction becomes stronger as the trait and hypothesis become more distinctive.

C Application Details

C.1 Experience cues

We construct 32 binary respondent-level indicators of experiences from five SCE modules: the monthly core survey, the credit-access module, the labor-market module, the housing module, and the spending module. Each cue is defined as an indicator for a specific SCE response option to a specific SCE question. Table D.I lists each cue and its SCE question wording together with the response that defines the indicator.

Table D.I: 32 SCE experience cues.

Cue	Survey question and defining response
became unemp.	Was working (full-time or part-time) in the previous wave of the survey and is no longer working in the current wave (any non-working status: not working, temporarily laid off, on sick leave, permanently disabled, retired, student, homemaker, or other).
became emp.	Was not working in the previous wave of the survey and is now working (full-time or part-time) in the current wave.
salary up	Their reported current annual earnings (before taxes and other deductions, including bonuses, overtime, tips, and commissions) increased by more than 5% (in log change) between consecutive survey waves.
salary down	Their reported current annual earnings (before taxes and other deductions, including bonuses, overtime, tips, and commissions) decreased by more than 5% (in log change) between consecutive survey waves.
changed employer	“Were you working in the same job when you submitted your last survey?” Answered: “No, I work for a different employer now”.
got married	In the previous wave of the survey reported not being currently married or living as a partner with someone; in the current wave reports being currently married or living as a partner with someone.
new young child	The number of children age 5 or younger living in their primary residence increased between two consecutive waves of the survey (e.g., a birth, adoption, or new young child joining the household).
moved	“Since [last survey], have you moved to a different primary residence (the place where you usually live)?” Answered: Yes.
credit reject	During the last twelve months, applied for a credit card, mortgage or home-based loan, auto loan, or student loan, and the request was rejected.
sick leave	“What is your current employment situation?” Answered: “On sick or other leave” or “Permanently disabled or unable to work”.
mort. missed	“During the past 12 months, did you miss two or more payments on any outstanding loans against your primary residence (including all mortgages and home equity loans)?” Answered: Yes.

Cue	Survey question and defining response
rent missed	"Were you unable to make a rent payment over the past 12 months?" Answered: Yes.
mort. refi	Refinanced their mortgage, and at the most recent refinance the interest rate was lowered.
buy: car	"Did any members of your household (including you) make any of the following large purchases during the last 4 months?" Selected: "Car or other vehicles".
buy: reno	"Did any members of your household (including you) make any of the following large purchases during the last 4 months?" Selected: "Home repairs, improvements or renovations".
buy: vacation	"Did any members of your household (including you) make any of the following large purchases during the last 4 months?" Selected: "Trips and vacations".
home buy	Purchased a house or apartment in the last 4 months and currently owns their primary residence.
job offer	"How many job offers did you receive in the last 4 months? Remember a job offer is not necessarily a job that you accepted." Answered: at least one.
job search	"Have you done anything in the LAST 4 WEEKS to look for new work?" Answered: "Yes, looking to possibly leave my current job for a new job".
fin better	"Do you think you (and any family living with you) are financially better or worse off these days than you were 12 months ago?" Answered: "Somewhat better off" or "Much better off".
credit max	"During the past 12 months, were you maxed out on at least one credit card at any time?" Answered: Yes.
loan late 30	"During the past 12 months, were you 30 days or more late on any loan payments?" Answered: Yes.
loan late 90	"During the past 12 months, were you 90 days or more late on any loan payments?" Answered: Yes.
credit tight	"During the past 12 months, did a lender close one of your credit accounts or lower your credit limit without your request?" Answered: Yes.
discouraged	"During the past 12 months, was there a time when you needed credit but did not apply because you thought you would be rejected?" Answered: Yes.
fin worse	"Do you think you (and any family living with you) are financially better or worse off these days than you were 12 months ago?" Answered: "Somewhat worse off" or "Much worse off".
credit OK	During the last twelve months, applied for a credit card, mortgage or home-based loan, auto loan, or student loan, and had the request fully granted.
buy: appliances	"Did any members of your household (including you) make any of the following large purchases during the last 4 months?" Selected: "Home appliances".
buy: electronics	"Did any members of your household (including you) make any of the following large purchases during the last 4 months?" Selected: "Electronics, computers or cell phones".
buy: furniture	"Did any members of your household (including you) make any of the following large purchases during the last 4 months?" Selected: "Furniture".

Cue	Survey question and defining response
good health	“Would you say your health is excellent, very good, good, fair, or poor?” Answered: “Excellent” or “Very good”.
poor health	“Would you say your health is excellent, very good, good, fair, or poor?” Answered: “Fair” or “Poor”.

C.2 Forecast outcomes and historical analogs

The dependent variables in our analysis are 16 SCE forecast questions. Each SCE forecast question asks the respondent for a numerical belief about a specific macroeconomic outcome over a specific horizon (e.g., the probability that stock prices are higher one year from now; the expected year-over-year rate of inflation). For each such forecast outcome we identify a historical macroeconomic series that is its natural ground-truth analog (e.g., the realized year-over-year return on the S&P 500 for stocks; realized CPI year-over-year inflation for inflation). This historical series, observed at annual frequency from 1961 to 2023, plays two roles in our analysis. First, it defines ψ_o : an LLM is asked how similar a respondent’s experience is to a one-year period in U.S. history, and the realized value of the historical analog in that period is the value of ψ_o in that “state” ω_y . Second, the historical series therefore defines $\mathcal{D}^{\psi_o}(c_j)$, the representativeness of cue j for outcome o . Table D.II lists, for each of the 16 forecast outcomes, the SCE question wording and its corresponding historical series.

Table D.II: 16 SCE macroeconomic expectations.

Outcome	SCE question	Historical analog
Stocks (1y direction)	“What do you think is the percent chance that 12 months from now, on average, stock prices in the U.S. stock market will be higher than they are now?”	$\mathbb{1}(\text{S\&P 500 annual return} > 0)$
Inflation (1y)	Probability-density mean of the respondent’s reported distribution over inflation/deflation rates over the next 12 months (10 bins).	Realized CPI YoY (%)
Unemployment (1y direction)	“What do you think is the percent chance that 12 months from now the unemployment rate in the U.S. will be higher than it is now?”	$\mathbb{1}(\Delta \text{unemployment rate} > 0)$
Interest rates (1y direction)	“What do you think is the percent chance that 12 months from now the average interest rate on savings accounts will be higher than it is now?”	$\mathbb{1}(\Delta \text{federal funds rate} > 0)$

Outcome	SCE question	Historical analog
Home prices (1y)	“By about what percent do you expect the average home price to [increase/decrease] by? Please give your best guess.”	Shiller nominal home-price index YoY (%)
Credit ease (1y)	“Do you think that 12 months from now it will generally be harder or easier for people to obtain credit or loans (including credit and retail cards, auto loans, student loans, and mortgages) than it is these days?” 1–5 Likert (1=Much harder, 5=Much easier).	–SLOOS net % of banks tightening standards on consumer loans
Inflation (3y)	Probability-density mean of the respondent’s reported distribution over inflation/deflation rates over the 12-month window 24–36 months after the survey.	Realized CPI 3-year trailing average YoY (%)
Inflation (5y)	Probability-density mean of the respondent’s reported distribution over inflation/deflation rates over the 12-month window 48–60 months after the survey.	Realized CPI 5-year trailing average YoY (%)
Mortgage rate (1y level)	“What do you think is the average interest rate on a new 30-year fixed-rate mortgage one year from today?” Level in %.	Freddie Mac 30-year fixed mortgage rate, annual average (%)
Mortgage rate (3y level)	“And three years from now, what do you think the average rate on a new 30-year fixed-rate mortgage would be?” Level in %.	Freddie Mac 30-year fixed mortgage rate, annual average (%)
Gov. debt (1y)	“By about what percent do you expect the level of U.S. government debt to [increase/decrease] by?”	Federal debt held by the public YoY (%)
Gas price (1y)	“Twelve months from now, what do you think will have happened to the price of a gallon of gas? Increased by __ % or decreased by __ %.”	BLS gasoline-component CPI YoY (%)
Food price (1y)	“Twelve months from now, what do you think will have happened to the price of food? Increased by __ % or decreased by __ %.”	BLS food-component CPI YoY (%)
Medical price (1y)	“Twelve months from now, what do you think will have happened to the price of medical care? Increased by __ % or decreased by __ %.”	BLS medical-care-component CPI YoY (%)
College price (1y)	“Twelve months from now, what do you think will have happened to the cost of a college education? Increased by __ % or decreased by __ %.”	BLS college-tuition-component CPI YoY (%)
Rent price (1y)	“Twelve months from now, what do you think will have happened to the cost of renting a typical house/apartment? Increased by __ % or decreased by __ %.”	BLS rent-of-primary-residence CPI YoY (%)

C.3 LLM similarity prompt

Each of the seven raters is sent the same prompt structure: a system prompt defining the task and the response format, and a single user message that describes the SCE question and response defining the cue as well as a bullet list of macroeconomic indicators for the target year. Each response is a JSON object {"similarity": <0-100 integer>, "reason": "<one sentence>"}. **System message:**

Rate how similar a person's economic situation is to the macroeconomic conditions of a given year in the U.S., on a 0-100 scale (0 = not similar, 100 = very similar).

Respond with JSON: {"similarity": <0-100 integer>, "reason": "<one sentence>"}

User message template (placeholders in italics):

Person's situation: *{{experience question text}}*

Macroeconomic conditions of *{{year}}* in the U.S.:
{{bullet list of macro indicators for that year}}

Rate the similarity 0-100 with a one-sentence reason.

Rendered example (experience = became unemp., year = 2008):

Person's situation: was working (full-time or part-time) in the previous wave of the survey; is no longer working in the current wave (e.g., answered "What is your current employment situation?" with "Not working, but would like to work" or "Temporarily laid off").

Macroeconomic conditions of 2008 in the U.S.:
- S&P 500 annual return: -37.0%
- Consumer price growth (CPI YoY): +3.9%
- Unemployment rate: 5.8%
- GDP growth: -0.1%
- 30-year mortgage rate: 6.0%
- Federal funds rate: 1.93%
- Home price index YoY: -12.0%
- Bank credit tightening (net % of banks): +41.9%
- Consumer sentiment index: 64

Rate the similarity 0-100 with a one-sentence reason.

C.4 Completeness and restrictiveness procedure

For both quantities, the unit of analysis is the (respondent $\times$ survey $\times$ belief) triple (2.3M obs) and the dependent variable is the SCE belief standardized within outcome. We run 5-fold cross-validation across respondents, so all rows of a given respondent live in

the same fold. The model is then refit on each training fold (grid search over W_C , closed-form $\mathcal{D}^{\psi_o}(C)$); we benchmark against a random forest with the 32 experience indicators and 16 outcome dummies as features (50 trees, `min_samples_leaf`=10). Completeness is

$$\text{Completeness} = \frac{\text{RMSE}_{\text{base}}^2 - \text{RMSE}_{\text{model}}^2}{\text{RMSE}_{\text{base}}^2 - \text{RMSE}_{\text{rf}}^2}.$$

Restrictiveness is computed via 20 random permutations σ of the 32 experience slots: for each σ we reassign the $(\mathcal{W}(c_j), \mathcal{D}^{\psi_o}(c_j))$ bundle so that “became unemp.” inherits cue $\sigma(\text{became unemp.})$ ’s bundle for all 16 outcomes *uniformly across respondents*, refit the structural model, and compute $\text{RMSE}_{\text{null}}$. Then

$$\text{Restrictiveness} = \overline{\text{RMSE}_{\text{null}}^2} / \text{RMSE}_{\text{base}}^2.$$